Sri Lanka Institute of Information Technology

Data warehousing and Business Intelligence

Assignment 1

WIDANAGAMAGE S S S – IT23641006

Year 03 Semester 01 - WE 02.01

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1.Introduction

This assignment outlines the design and implementation of a Retail Data Warehouse using SQL Server Management Studio (SSMS) and SQL Server Integration Services (SSIS). The selected retail dataset had information on customers, orders, products, shipping information, payment information, feedback, ratings, and sales values from a single source. In order to meet the requirements of this assignment and apply data warehousing concepts, the dataset was prepared and divided into three different source types from a single source. This prepared data was then normalized and implemented as a dimensioned data warehouse.

Business scenario chosen for this assignment**: Retail Sales Analysis.**

Data warehousing concepts implemented:

- Preparation of the source from a single source
- Use of three different source types
- Normalization during source preparation
- Design of the dimension and fact tables
- Implementation using SQL Server Integration Services
- Implementation of Slowly Changing Dimension

2.Original Dataset Description

The original retail dataset was provided as a single combined dataset containing attributes from multiple business subjects. It included customer information, transaction information, product information, and order profile information in one source.

Transaction_	Customer_ID	Name	Email	Phone	Address	City	State	Zipcode	Country	Age	Gender	Income	Customer_S	Date	Year	Month	Time	Totsl_Purch	Amount	Totsl_Amou	Product_Cat	Product_Bra	Product_Typ	Product_Co	Feedback	Shipping_Me	Payment_Me	Order_Status	Ratings	products
8691788	37249	Michelle Harr	Ebony39@g	1.415E+09	3959 Amand	Dortmund	Berlin	77985	Germany	21	Male	Low	Regular	9/18/2023	2023	September	22:03:55	3	108.02876	324.08627	Clothing	Nike	Shorts	Clothing|Nik	Excellent	Same-Day	Debit Card	Shipped	5	Cycling shorts
2174773	63743	Kelsey Hill	Mark36@gr	6.653E+09	82072 Dawn	Nottingham	England	39071	UK	19	Female	Low	Premium	12/31/2023	2023	December	8:42:04	2	400.35391	806.70781	Electronics	Samsung	Tablet	Electronics|S	Excellent	Standard	Credit Card	Processing	4	Lenovo Tab
6679610	30192	Scott Jensen	Shaun85@gr	8.362E+09	4133 Young	Geelong	New South W	75329	Australia	48	Male	Low	Regular	4/26/2023	2023	April	4:06:23	3	354.4776	1063.4328	Books	Penguin Boo	Children's	Books|Peng	Average	Same-Day	Credit Card	Processing	2	Sports equipment
7232460	62101	Joseph Mills	Mary34@gr	2.777E+09	8148 Thomas	Edmonton	Ontario	88420	Canada	56	Male	High	Premium	5/8/2023	2023	May	14:55:17	7	352.40712	2466.854	Home Decor	Home Depot	Tools	Home Decor|	Excellent	Standard	PayPal	Processing	4	Utility knife
4383775	27301	Debra Colem	Charles30@	9.098E+09	5813 Lori Pa	Bristol	England	48704	UK	22	Male	Low	Premium	1/10/2024	2024	January	16:54:07	2	124.27652	248.55305	Grocery	Nestle	Chocolate	Grocery|Nest	Bad	Standard	Cash	Shipped	1	Chocolate cookies
6095326	41283	Ryan Johnson	Haley12@gm	3.283E+09	532 Ashley C	Brisbane	New South W	74430	Australia	59	Female	Medium	Premium	9/21/2023	2023	September	23:24:27	4	296.23981	1185.1672	Electronics	Apple	Tablet	Electronics|A	Good	Express	PayPal	Pending	4	Lenovo Tab
5434096	37285	Erin Lewis	Arthur76@gr	1.578E+09	600 Brian Pa	Kitchener	Ontario	47545	Canada	29	Female	Low	New	6/26/2023	2023	June	13:35:51	2	315.05765	630.1153	Electronics	Samsung	Television	Electronics|S	Bad	Standard	Cash	Processing	1	OLED TV
2344675	26600	Angela Field	Tanya34@gr	3.666E+09	237 Young C	Munich	Berlin	86862	Germany	29	Male	Medium	Premium	3/24/2023	2023	March	10:12:56	1	46.58807	46.58807	Clothing	Zara	Shirt	Clothing|Zar	Bad	Same-Day	Cash	Processing	1	Dress shirt
4155845	80115	Diane Clark	Martin39@g	6.22E+09	8823 Marish	Wollongong	New South W	39820	Australia	46	Male	Medium	New	1/6/2024	2024	January	14:38:26	8	328.8393	2630.7144	Grocery	Nestle	Chocolate	Grocery|Nest	Bad	Same-Day	Cash	Delivered	1	Dark chocolate
4326148	31878	Lori Ball	Jessica33@g	6.005E+09	6225 William	Cologne	Berlin	64311	Germany	25	Male	Medium	New	10/4/2023	2023	October	22:27:40	10	397.61823	3976.1823	Home Decor	Home Depot	Decorations	Home Decor|	Excellent	Standard	Cash	Delivered	4	Candles
8493213	19136	Jonathan Est	Mark38@gr	2.397E+09	9772 Sara C	Portsmouth	England	59280	UK	64	Female	Low	Regular	7/20/2023	2023	July	23:06:51	4	90.98187	363.92748	Home Decor	Home Depot	Tools	Home Decor|	Average	Standard	Credit Card	Shipped	2	Screwdriver set
1609659	66883	Brianna Ocon	David47@gm	9.398E+09	95411 Jerry P	Portsmouth	England	31253	UK	31	Female	Medium	Regular	6/21/2023	2023	June	13:09:58	1	364.83057	364.83057	Books	Random Hou	Non-Fiction	Books|Rand	Average	Standard	Credit Card	Pending	2	Science
2846632	31930	Kristina Willi	Charles23@	1.623E+09	60629 Jonas	Portsmouth	England	38173	UK	56	Female	Medium	Regular	1/2/2024	2024	January	0:00:47	10	161.87836	1618.7836	Grocery	Coca-Cola	Water	Grocery|Coc	Bad	Standard	PayPal	Delivered	1	Bottled water
3211461	74671	Felicia Brown	Laura72@gm	8.87E+09	381 Richard	Portsmouth	England	32061	UK	53	Female	Medium	Regular	5/7/2023	2023	May	18:49:35	3	350.81702	3151.3531	Grocery	Nestle	Snacks	Grocery|Nest	Excellent	Express	PayPal	Delivered	4	Fruit snacks
2401331	96300	Andrew Guzr	Eric76@gma	2.923E+09	470 Rodrigo	Portsmouth	England	5259	UK	32	Female	High	Regular	11/18/2023	2023	November	23:41:05	6	297.72604	1786.3562	Clothing	Adidas	T-shirt	Clothing|Adi	Bad	Same-Day	Cash	Shipped	1	V-neck tee
4363987	64395	Justin Lara	Jacob26@gr	2.683E+09	3577 Cole St	Portsmouth	England	9845	UK	43	Male	Medium	Regular	6/15/2023	2023	June	23:24:22	3	23.723901	71.171704	Books	Random Hou	Literature	Books|Rand	Bad	Express	Credit Card	Pending	1	Drama
6661000	63909	Jessica Harris	Anna45@gm	2.54E+09	93806 Mary	Portsmouth	England	84007	UK	63	Male	Low	Regular	7/1/2023	2023	July	11:12:02	10	14.127164	141.27164	Grocery	Pepsi	Water	Grocery|Peps	Average	Same-Day	Debit Card	Shipped	2	Flavored water
2430572	71803	Alicia Knight	Jennifer16@gr	6.262E+09	9656 Moyers	Portsmouth	England	91367	UK	19	Female	High	Regular	4/14/2023	2023	April	5:50:55	3	139.78686	419.36057	Electronics	Apple	Tablet	Electronics|A	Good	Express	Cash	Shipped	3	Samsung Galaxy Tab
7128039	78376	Ronald Chen	Nicole72@gr	1.786E+09	6048 Charlie	Portsmouth	England	82975	UK	49	Male	Low	Regular	2/7/2024	2024	February	18:29:59	1	210.19792	210.19792	Grocery	Coca-Cola	Juice	Grocery|Coc	Bad	Express	Credit Card	Shipped	1	Orange juice
3489336	40654	Sara Rogers	Jeff39@gma	2.668E+09	6048 Charlie	Portsmouth	England	3035	UK	63	Male	Low	Regular	10/24/2023	2023	October	13:01:04	2	469.65964	939.31928	Home Decor	IKEA	Furniture	Home Decor|	Average	Standard	Cash	Shipped	2	Bookshelf
6325572	42165	Daniel Hatch	Oscar54@gm	3.604E+09	37402 Willia	Portsmouth	England	29119	UK	46	Male	Low	Regular	3/21/2023	2023	March	11:51:41	6	420.87503	2525.2502	Grocery	Nestle	Coffee	Grocery|Nest	Average	Same-Day	PayPal	Processing	2	Affogato
8871554	63914	Joshua Nobl	Diana25@gr	5.383E+09	7467 Henry P	Portsmouth	England	16385	UK	61	Male	High	Regular	8/6/2023	2023	August	5:13:31	4	200.28713	801.14851	Books	HarperCollin	Non-Fiction	Books|Harpe	Average	Standard	PayPal	Processing	2	Business
8476855	42410	Anthony Roc	Paula43@gr	3.745E+09	585 Malik Lo	Portsmouth	England	83803	UK	21	Female	Low	Regular	2/3/2024	2024	February	6:22:00	6	34.124266	204.7456	Books	Penguin Boo	Non-Fiction	Books|Pengu	Average	Same-Day	PayPal	Processing	2	Self-help
9338306	48863	Ashley Waterhouse	Christian23@	3.296E+09	6088 Sandra	Portsmouth	England	38435	UK	41	Female	Low	Regular	5/31/2023	2023	May	9:15:44	8	436.60731	3492.8633	Home Decor	Bed Bath & E	Bathroom	Home Decor|	Good	Standard	PayPal	Pending	3	Sink
3070447	93670	Timothy Bent	Thomas42@g	4.755E+09	424 Patton F	Portsmouth	England	2916	UK	38	Male	Medium	Regular	2/27/2024	2024	February	13:51:11	2	285.29833	570.59666	Grocery	Coca-Cola	Juice	Grocery|Coc	Bad	Same-Day	PayPal	Processing	1	Grape juice
9082305	47037	Mike Christe	Olivia35@gr	3.606E+09	753 Rachel L	Portsmouth	England	92862	UK	59	Male	High	Regular	4/20/2023	2023	April	4:45:57	6	75.209935	451.25961	Home Decor	Bed Bath & E	Kitchen	Home Decor|	Bad	Standard	Debit Card	Pending	1	Stove
4725198	28381	Kimberly Erv	Chris79@gr	8.215E+09	2445 Clayton	Portsmouth	England	30007	UK	20	Male	High	Regular	6/3/2023	2023	June	4:57:04	1	173.57074	173.57074	Electronics	Samsung	Smartphone	Electronics|S	Good	Express	PayPal	Shipped	3	Huawei P
5230876	23097	Michael Cole	Rachel75@gr	6.487E+09	0777 Rebecc	Portsmouth	England	14668	UK	67	Male	Low	Regular	1/7/2024	2024	January	17:33:45	5	471.48213	2357.4107	Electronics	Apple	Tablet	Electronics|A	Bad	Same-Day	Credit Card	Shipped	1	Amazon Fire Tablet
2459574	42540	Robin Jackso	James3@gm	4.258E+09	8306 Daniel	Portsmouth	England	62300	UK	43	Male	Medium	Regular	4/22/2023	2023	April	9:17:29	8	333.85966	2670.8773	Home Decor	Home Depot	Decorations	Home Decor|	Excellent	Standard	Cash	Delivered	5	Curtains
2873384	55388	Yolanda Mur	Emily34@gr	2.451E+09	62142 Karen	Portsmouth	England	63033	UK	50	Male	Medium	Regular	6/21/2023	2023	June	16:42:37	6	240.22913	1441.3691	Grocery	Coca-Cola	Juice	Grocery|Coc	Good	Standard	Cash	Pending	3	Pomegranate juice
3413236	98256	Steven Boyd	Sheryl19@gr	5.243E+09	1713 Christo	Portsmouth	England	43602	UK	26	Female	Low	Regular	2/6/2024	2024	February	1:59:20	3	159.30419	479.41258	Clothing	Adidas	Shoes	Clothing|Adi	Good	Same-Day	Debit Card	Pending	3	Running shoes
6408632	90335	Raymond Av	Christopher4	4.239E+09	374 Snyder F	Portsmouth	England	34428	UK	56	Female	Low	Regular	11/26/2023	2023	November	6:43:16	2	147.03823	294.13647	Books	HarperCollin	Thriller	Books|Harpe	Good	Standard	Debit Card	Processing	3	Political thriller
8973612	65701	Christopher I	Derrick51@g	9.286E+09	57136 Martha	Portsmouth	England	11531	UK	66	Male	Medium	Regular	5/22/2023	2023	May	11:24:20	7	344.67919	2412.7543	Home Decor	IKEA	Decorations	Home Decor|	Excellent	Same-Day	Debit Card	Processing	5	Curtains
5404167	23713	Brian Gonzal	Steven34@g	4.686E+09	861 Beth Bra	Portsmouth	England	55397	UK	31	Male	Low	Regular	6/28/2023	2023	June	11:35:43	4	243.64028	974.56111	Electronics	Samsung	Smartphone	Electronics|S	Average	Express	Cash	Pending	2	Google Pixel
4208526	66911	Anne Doming	Christian61@	3.655E+09	704 Steven P	Portsmouth	England	71394	UK	24	Female	Low	Regular	12/16/2023	2023	December	4:00:08	2	333.70443	667.40887	Grocery	Nestle	Chocolate	Grocery|Nest	Excellent	Same-Day	Credit Card	Pending	4	Chocolate mousse
1505270	38517	Todd Jackso	Daniel66@gr	6.02E+09	85473 Palme	Portsmouth	England	43394	UK	54	Male	Low	Regular	4/10/2023	2023	April	6:50:20	3	322.14771	966.44314	Home Decor	IKEA	Decorations	Home Decor|	Good	Same-Day	PayPal	Delivered	4	Vases
8763109	65305	Jonathan Jac	Christian13@	8.524E+09	890 Brooke	Portsmouth	England	68300	UK	19	Male	High	Regular	12/12/2023	2023	December	17:48:31	5	135.1514	377.15701	Clothing	Zara	Shirt	Clothing|Zar	Good	Express	Credit Card	Pending	4	Henley shirt
4578663	31767	Brittany Rod	Lindsay16@g	3.325E+09	61440 Andre	Portsmouth	England	1527	UK	28	Male	High	Regular	2/16/2024	2024	February	20:44:42	9	161.38733	1452.486	Books	Penguin Boo	Children's	Books|Pengu	Excellent	Standard	Cash	Pending	4	Toys
5392046	30637	Mike Orozco	Kimberly48@	8.321E+09	8887 Alexan	Portsmouth	England	79514	UK	34	Female	High	Regular	2/8/2024	2024	February	12:07:52	7	44.330184	310.31129	Grocery	Pepsi	Soft Drink	Grocery|Peps	Good	Express	PayPal	Processing	4	Iced tea
2458233	96840	Susan Thoma	Bonnie80@g	3.541E+09	48415 Fergu	Portsmouth	England	85074	UK	65	Male	Low	Regular	10/28/2023	2023	October	13:43:05	8	28.82511	230.60408	Clothing	Adidas	Shoes	Clothing|Adi	Excellent	Same-Day	Credit Card	Pending	4	Running shoes
2055036	25848	Emma Palmer	Lisa7@gmail.	8.201E+09	416 Angel M	Portsmouth	England	42516	UK	40	Male	High	Regular	2/23/2024	2024	February	22:28:09	10	20.528537	205.28537	Grocery	Coca-Cola	Juice	Grocery|Coc	Excellent	Same-Day	PayPal	Pending	4	Apple juice
4323734	31112	Amanda Dod	Heather32@g	8.188E+09	563 James H	Portsmouth	England	83908	UK	64	Female	High	Regular	3/11/2023	2023	March	8:03:15	3	227.42555	682.27665	Books	Random Hou	Literature	Books|Rand	Good	Same-Day	Debit Card	Pending	4	Drama
3902930	77572	Joce Spencer	Amber67@g	6.282E+09	02936 Roy F	Portsmouth	England	36771	UK	68	Female	High	Regular	10/6/2023	2023	October	4:59:02	6	334.1657	2004.9342	Electronics	Apple	Laptop	Electronics|A	Bad	Same-Day	Debit Card	Shipped	1	Microsoft Surface Laptop
9885563	46361	Priscilla Wig	Patricia51@g	3.836E+09	0430 Moody	Portsmouth	England	35203	UK	46	Male	High	Regular	11/5/2023	2023	November	17:36:05	2	141.21437	282.42873	Home Decor	IKEA	Furniture	Home Decor|	Excellent	Same-Day	PayPal	Processing	5	Desk
3239367	62464	Michele John	Melissa70@	1.347E+09	622 Diana Ci	Portsmouth	England	15160	UK	58	Female	High	Regular	12/14/2023	2023	December	20:24:39	3	312.97657	938.92972	Clothing	Zara	Dress	Clothing|Zar	Average	Standard	Cash	Delivered	2	Bodycon dress
6047327	74036	Robert Warr	Hannah73@g	1.224E+09	489 Karr Isla	Portsmouth	England	12030	UK	36	Male	Low	Regular	6/19/2023	2023	June	11:40:40	1	399.78712	399.78712	Electronics	Sony	Headphones	Electronics|S	Good	Express	Credit Card	Pending	4	Sports headphones
1069259	21399	Mario Smith	Joshua63@g	2.207E+09	192 Kimberly	Portsmouth	England	85340	UK	57	Male	High	Regular	3/16/2023	2023	March	1:42:10	7	315.62887	2209.4021	Home Decor	IKEA	Lighting	Home Decor|	Excellent	Same-Day	Cash	Pending	5	Pendant lights
6353946	33757	Luana Hayes	Haley86@gm	5.845E+09	726 Eaton St	Portsmouth	England	58226	UK	27	Female	Medium	Regular	7/16/2023	2023	July	17:20:37	8	93.06219	744.05375	Home Decor	Home Depot	Decorations	Home Decor|	Good	Same-Day	Credit Card	Processing	3	Clocks
8786466	90189	Maria Tucker	Amber14@g	1.368E+09	34455 Foste	Portsmouth	England	44399	UK	35	Female	Medium	Regular	4/19/2023	2023	April	2:13:39	7	346.71959	2427.0371	Home Decor	Bed Bath & E	Bedding	Home Decor|	Bad	Standard	Credit Card	Delivered	1	Comforter
2545225	86515	Derek Green	Sara74@gma	8.38E+09	33744 April	Portsmouth	England	14086	UK	57	Female	Medium	Regular	9/10/2023	2023	September	19:22:43	6	106.50674	639.04045	Clothing	Adidas	Jacket	Clothing|Adi	Good	Standard	Credit Card	Pending	2	Peacoat
9051430	19303	Rachel Hobb	Daniel23@gr	4.798E+09	50819 Moyer	Portsmouth	England	30346	UK	36	Male	Low	Regular	3/15/2023	2023	March	1:07:58	2	230.8985	461.7197	Home Decor	Home Depot	Furniture	Home Decor|	Bad	Standard	PayPal	Processing	5	Bed
4580441	12911	Chelsea Cum	Daniel35@gr	3.051E+09	336 Sara Ra	Portsmouth	England	27283	UK	65	Male	High	Regular	11/11/2023	2023	November	15:38:31	3	394.60627	1183.8188	Grocery	Nestle	Chocolate	Grocery|Nest	Excellent	Express	PayPal	Processing	5	White chocolate
9786592	71257	Bryan Edwar	Gregory3@g	8.84E+09	816 Annette	Portsmouth	England	94747	UK	59	Male	High	Regular	7/25/2023	2023	July	20:29:30	2	281.60017	563.20034	Electronics	Samsung	Television	Electronics|S	Average	Standard	PayPal	Delivered	2	Android TV
6315151	75456	Christopher I	Tonya57@gr	7.162E+09	707 Michael	Portsmouth	England	87547	UK	38	Female	High	Regular	10/3/2023	2023	October	4:34:26	4	303.76386	1215.1535	Electronics	Sony	Headphones	Electronics|S	Bad	Express	Cash	Delivered	1	Studio headphones
6357160	50574	James Chard	Andrew57@g	8.004E+09	7645 Stewart	Portsmouth	England	52548	UK	46	Male	Low	Regular	10/17/2023	2023	October	6:40:06	4	129.75697	519.02789	Clothing	Nike	Shoes	Clothing|Nik	Bad	Same-Day	Debit Card	Shipped	1	Sneakers
8523222	81200	Tony Moon	Brandon73@	2.089E+09	28526 Sandr	Portsmouth	England	15193	UK	24	Male	Medium	Regular	6/27/2023	2023	June	11:30:53	4	257.58697	1030.3479	Electronics	Sony	Headphones	Electronics|S	Good	Same-Day	Cash	Processing	4	Studio headphones
6321233	21381	Stanley Guier	April61@gm	5.514E+09	0511 Barnard	Portsmouth	England	36935	UK	67	Male	Medium	Regular	11/26/2023	2023	November	16:03:33	10	387.55345	3875.5345	Home Decor	Home Depot	Decorations	Home Decor|	Average	Same-Day	Debit Card	Shipped	2	Sculptures
9658399	71121	Heather Stoa	Taylor73@gr	6.417E+09	55047 Ashley	Portsmouth	England	60085	UK	54	Female	High	Regular	2/18/2024	2024	February	11:46:03	10	97.936789	979.36789	Grocery	Pepsi	Soft Drink	Grocery|Peps	Bad	Same-Day	Cash	Shipped	1	Cream soda
3121715	64722	Richard Mor	Elizabeth40@	3.732E+09	55900 Cassa	Portsmouth	England	56647	UK	68	Male	Medium	Regular	2/22/2024	2024	February	19:13:49	9	291.88127	2626.3315	Books	Random Hou	Literature	Books|Rand	Average	Express	Credit Card	Shipped	2	Drama
4021565	53619	Christina Sne	Wendy95@g	1.468E+09	01232 Austin	Portsmouth	England	28089	UK	29	Male	Medium	Regular	10/8/2023	2023	October	5:45:56	9	474.82694	4273.4425	Books	Random Hou	Non-Fiction	Books|Rand	Average	Standard	PayPal	Shipped	2	History

Because the original file mixed different business subjects in one source, it was not suitable for direct warehouse loading without preparation. Therefore, source preparation and normalization were performed before ETL development.

3.Preparation of Data Sources

3.1 Source Preparation Strategy

The original retail dataset was normalized during source preparation by separating the combined data into three independent business subjects: order/transaction data, customer master data, and product master data. Instead of keeping all descriptive and transactional attributes in one source, customer-related attributes were extracted into Customer_Master.csv, product-related attributes were prepared in Product_Master.xlsx, and order/transaction-related attributes were loaded into the SQL Server source table SrcRetailOrders. This normalization reduced redundancy, avoided repeated storage of customer and product descriptions in every transaction row, and created a clearer separation of business entities before ETL processing. As a result, the prepared sources were easier to manage, better aligned with warehouse dimensional loading, and more suitable for integrating multiple source types in the staging and data warehouse environment.

The prepared sources were:

- **Database Source**: order and transaction data
- **CSV Source**: customer master data
- **Excel Source**: product master/hierarchy data

3.2 Sources Used

Source 1 – Database Source

A SQL Server table was created to hold the normalized retail order data:

Retail_SourceDB2.dbo.SrcRetailOrders

This source contains:

- transaction identifiers
- customer identifiers
- product codes
- transaction dates and times
- quantities and sales amounts
- shipping, payment, status, feedback, and ratings

Source 2 – CSV Source

A customer master file was prepared in CSV format:

Customer_Master.csv

This source contains:

- customer identifier
- personal details
- geographic details
- demographic attributes
- segmentation attributes

Source 3 – Excel Source

A product master file was prepared in Excel format:

Product_Master.xlsx

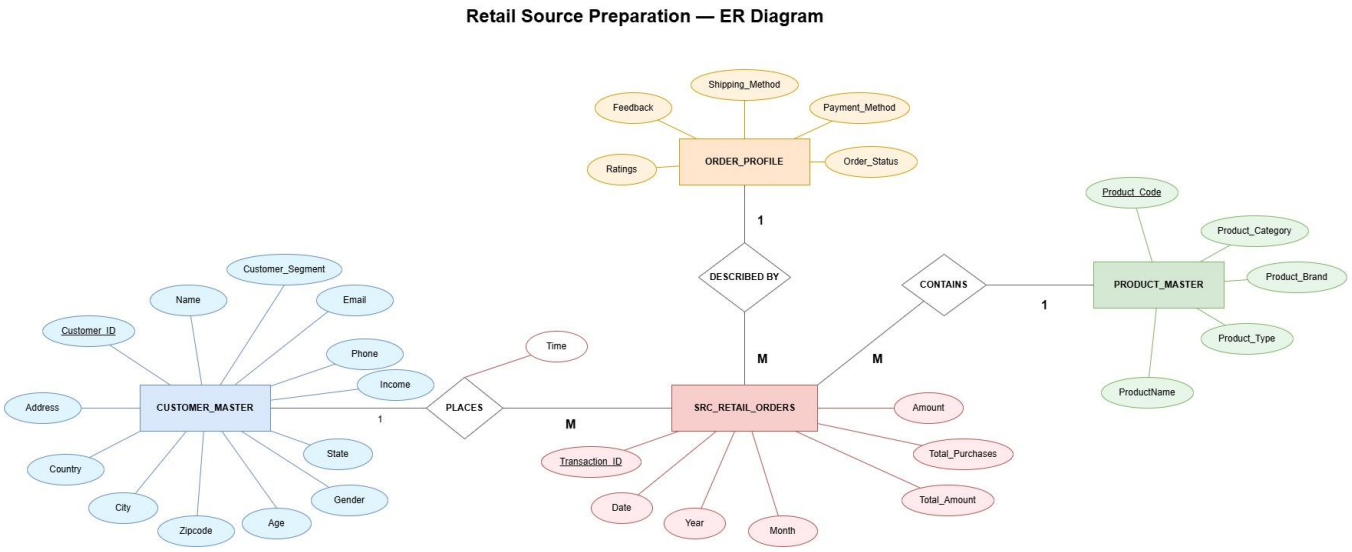
This source contains:

- product code
- product category
- product brand
- product type
- product name

3.3 Attributes Available in Each Source

Source Name	Format	Key Attributes
SrcRetailOrders	SQL Server table	Transaction_ID Customer_ID Product_Code Date Year Month Time Total_Purchases Amount Total_Amount Feedback Shipping_Method Payment_Method Order_Status Ratings
Customer_Master.csv	CSV	Customer_ID Name Email Phone Address City State Zipcode Country Age Gender Income Customer_Segment
Product_Master.xlsx	Excel	Product_Code Product_Category Product_Brand Product_Type ProductName

3.4 Entity Relationship Diagram



4. Database Environment and Tools Used

The following databases were created:

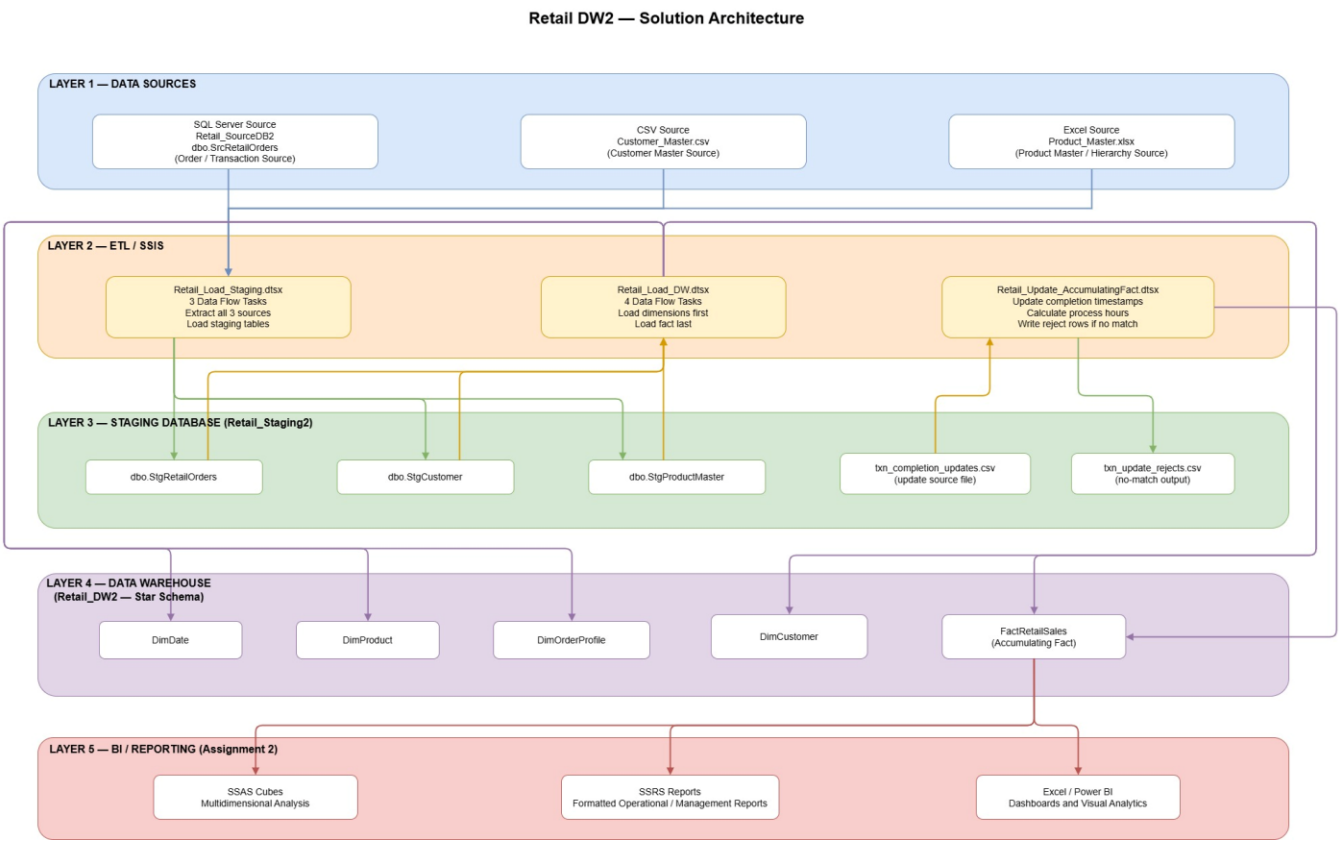
- Retail_SourceDB2
- Retail_Staging2
- Retail_DW2

The following tools were used:

- **SQL Server Management Studio (SSMS)** for database creation, table creation, stored procedures, and validation queries
- **Visual Studio with SSIS** for ETL package development

5. Solution Architecture

5.1 Architecture Diagram



Component Descriptions

1. Data Sources Layer

In this layer where the raw data is stored before ETL begin. In my project this layer include **three different source types**. A**SQL Server source table**(Retail_SourceDB2.dbo.SrcRetailOrders) is used to order and transaction data, a **CSV flat file source** (Customer_Master.csv) is used for customer data,and **Excel source file** (Product_Master.xlsx) is use for product and hierarchy data. In addition, separate CSV file (txn_completion_updates.csv) is used later to update completion details in the accumulating fact table.Data in this layer is kept in its prepared but still source-oriented form.

2. ETL Layer (SSIS)

SSIS is the middleware layer that connect the prepared data sources to the staging database and the data warehouse.In my project, SSIS extracts data from the SQL, CSV, and Excel sources, applies transformation logic such as data type conversion, lookup matching, filtering of invalid rows,and derived column generation, and then loads processed data into staging and warehouse tables. SSIS also handle the Slowly Changing Dimension processs for customer data and perform the later update of accumulating fact records using the completion update file.

3. Staging Database

The staging layer act as a temporary holding area for extracted source data before warehouse transformation. In my project, the staging database is Retail_Staging2, and its contains one staging table per main source StgRetailOrders, StgCustomer, and StgProductMaster. The staging layer separate raw source extraction from warehouse loading, making it easier to handle datatype issues, reload data safely, and debug the ETL process. The staging tables are truncated before each run so that the layer behaves as a clean reloadable buffer

4. Data Warehouse

The main analytical store is the data warehouse Retail_DW2. It is organize as **Star Schema** with one central fact table, FactRetailSales, connected to four dimension tables DimDate, DimCustomer, DimProduct, and DimOrderProfile. DimCustomer implement **Slowly Changing Dimension Type 2** logic for selected historical attributes so that important customer history can be preserved over time.The fact table is designed as **accumulating snapshot fact table**, where transaction creation time is loaded first and completion-related columns are updated later.

5. BI Reporting Layer

The BI reporting layer consum structured data from the data warehouse and presents it to businesss users through reports, dashboards, and analytical models. In this project, this layer is intended to support tools such as **SSAS** for multidimensional analysis, **SSRS** for formatted reports, and **Excel / Power BI** for dashboard-style analysis.

5.2 Staging Tables

	TableName	rowCount
1	StgRetailOrders	86753
2	StgCustomer	86753
3	StgProductMaster	592

StgRetailOrders

Stores extracted order and transaction data from the database source.

```
CREATE TABLE dbo.StgRetailOrders
(
  Transaction_ID     NVARCHAR(50)   NULL,
  Customer_ID        NVARCHAR(50)   NULL,
  Product_Code       NVARCHAR(255)  NULL,
  [Date]             DATE           NULL,
  [Year]             SMALLINT       NULL,
  [Month]            NVARCHAR(20)   NULL,
  [Time]             TIME(7)        NULL,
  Total_Purchases    INT            NULL,
  Amount             DECIMAL(18,2)  NULL,
  Total_Amount       DECIMAL(18,2)  NULL,
  Feedback           NVARCHAR(50)   NULL,
  Shipping_Method    NVARCHAR(50)   NULL,
  Payment_Method     NVARCHAR(50)   NULL,
  Order_Status       NVARCHAR(50)   NULL,
  Ratings            INT            NULL
);
```

	Transaction_ID	Customer_ID	Product_Code	Date	Year	Month	Time	Total_Purchases	Amount	Total_Amount	Feedback	Shipping_Method	Payment_Method	Order_Status	Ratings
1	8691788	37249	Clothing|Nike|Shorts|Cycling shorts	2023-09-18	2023	September	22:03:55.0000000	3	108.03	324.09	Excellent	Same-Day	Debit Card	Shipped	5
2	2174773	69749	Electronics|Samsung|Tablet|Lenovo Tab	2023-12-31	2023	December	08:42:04.0000000	2	403.35	806.71	Excellent	Standard	Credit Card	Processing	4
3	6679610	30192	Books|Penguin Books|Children's|Sports equipment	2023-04-26	2023	April	04:06:29.0000000	3	354.48	1063.43	Average	Same-Day	Credit Card	Processing	2
4	7232460	62101	Home Decor|Home Depot|Tools|Utility knife	2023-05-08	2023	May	14:55:17.0000000	7	352.41	2466.85	Excellent	Standard	PayPal	Processing	4
5	4983775	27901	Grocery|Nestle|Chocolate|Chocolate cookies	2024-01-10	2024	January	16:54:07.0000000	2	124.28	248.55	Bad	Standard	Cash	Shipped	1
6	6095326	41289	Electronics|Apple|Tablet|Lenovo Tab	2023-09-21	2023	September	23:24:27.0000000	4	296.29	1185.17	Good	Express	PayPal	Pending	4
7	5434096	97285	Electronics|Samsung|Television|QLED TV	2023-06-26	2023	June	13:35:51.0000000	2	315.06	630.12	Bad	Standard	Cash	Processing	1
8	2344675	26603	Clothing|Zara|Shirt|Dress shirt	2023-03-24	2023	March	10:12:56.0000000	1	46.59	46.59	Bad	Same-Day	Cash	Processing	1
9	4155845	80175	Grocery|Nestle|Chocolate|Dark chocolate	2024-01-06	2024	January	14:38:26.0000000	8	328.84	2630.71	Bad	Same-Day	Cash	Delivered	1
10	4926148	31878	Home Decor|Home Depot|Decorations|Candles	2023-10-04	2023	October	22:27:40.0000000	10	397.61	3976.11	Excellent	Standard	Cash	Delivered	4
11	8493213	19136	Home Decor|Home Depot|Tools|Screwdriver set	2023-07-20	2023	July	23:06:51.0000000	4	90.98	363.93	Average	Standard	Credit Card	Shipped	2
12	1609659	66883	Books|Random House|Non-Fiction|Science	2023-06-21	2023	June	13:09:58.0000000	1	364.83	364.83	Average	Standard	Credit Card	Pending	2
13	2846832	31930	Grocery|Coca-Cola|Water|Bottled water	2024-01-02	2024	January	00:00:47.0000000	10	161.88	1618.79	Bad	Standard	PayPal	Delivered	1
14	3217461	74671	Grocery|Nestle|Snacks|Fruit snacks	2023-05-07	2023	May	18:49:35.0000000	9	350.82	3157.35	Excellent	Express	PayPal	Delivered	4
15	2401331	98300	Clothing|Adidas|T-shirt|V-neck tee	2023-11-18	2023	November	23:41:05.0000000	6	297.73	1786.36	Bad	Same-Day	Cash	Shipped	1
16	4969987	64995	Books|Random House|Literature|Drama	2023-06-15	2023	June	23:24:22.0000000	3	23.72	71.17	Bad	Express	Credit Card	Pending	1
17	6681000	69939	Grocery|Pepsi|Water|Flavored water	2023-07-01	2023	July	11:12:02.0000000	10	14.73	147.27	Average	Same-Day	Debit Card	Shipped	2
18	2430572	71803	Electronics|Apple|Tablet|Samsung Galaxy Tab	2023-04-14	2023	April	05:50:55.0000000	3	139.79	419.36	Good	Express	Cash	Shipped	3
19	7728099	78376	Grocery|Coca-Cola|Juice|Orange juice	2024-02-07	2024	February	18:29:59.0000000	1	210.14	210.14	Bad	Express	Credit Card	Shipped	1
20	3489336	40654	Home Decor|IKEA|Furniture|Bookshelf	2023-10-24	2023	October	13:01:04.0000000	2	469.66	939.32	Average	Standard	Cash	Shipped	2
21	6925572	42165	Grocery|Nestle|Coffee|Affogato	2023-03-21	2023	March	11:51:41.0000000	6	420.88	2525.25	Average	Same-Day	PayPal	Processing	2
22	8871554	63914	Books|HarperCollins|Non-Fiction|Business	2023-08-06	2023	August	05:13:31.0000000	4	200.29	801.15	Average	Standard	PayPal	Processing	2
23	9476855	42410	Books|Penguin Books|Non-Fiction|Self-help	2024-02-03	2024	February	06:22:00.0000000	6	34.12	204.75	Average	Same-Day	PayPal	Processing	2
24	9338306	48869	Home Decor|Bed Bath & Beyond|Bathroom|Sink	2023-05-31	2023	May	09:15:44.0000000	8	436.61	3492.86	Good	Standard	PayPal	Pending	3
25	9070447	99870	Grocery|Coca-Cola|Juice|Grape juice	2024-02-27	2024	February	13:51:11.0000000	2	285.30	570.60	Bad	Same-Day	PayPal	Processing	1
26	9082305	47037	Home Decor|Bed Bath & Beyond|Kitchen|Stove	2023-04-20	2023	April	04:45:57.0000000	6	75.21	451.26	Bad	Standard	Debit Card	Pending	1

StgCustomer

Stores extracted customer master records from the CSV source.

```
CREATE TABLE dbo.StgCustomer
(
  Customer_ID      NVARCHAR(50)  NULL,
  Name             NVARCHAR(100) NULL,
  Email            NVARCHAR(100) NULL,
  Phone            NVARCHAR(50)  NULL,
  Address          NVARCHAR(200) NULL,
  City             NVARCHAR(50)  NULL,
  State            NVARCHAR(50)  NULL,
  Zipcode          NVARCHAR(50)  NULL,
  Country          NVARCHAR(50)  NULL,
  Age              NVARCHAR(50)  NULL,
  Gender           NVARCHAR(20)  NULL,
  Income           NVARCHAR(50)  NULL,
  Customer_Segment NVARCHAR(50)  NULL
);
```

	Customer_ID	Name	Email	Phone	Address	City	State	Zipcode	Country	Age	Gender	Income	Customer_Segment
1	72462	Christopher Snyder	Matthew18@gmail.com	1511799368	2960 Michael Gardens Apt. 389	Portsmouth	England	18095	UK	24	Male	Medium	Regular
2	18567	Kathryn Marshall MD	Joshua8@gmail.com	7574729332	602 Robert Locks Suite 414	Portsmouth	England	24677	UK	24	Male	Medium	Regular
3	92909	Alex Clark	Karen49@gmail.com	2934697623	414 Cassandra Freeway Apt. 574	Portsmouth	England	98298	UK	24	Male	Medium	Regular
4	51116	Curtis Sparks	Beth44@gmail.com	1739584671	620 Aaron Brook	Portsmouth	England	30174	UK	24	Male	Medium	Regular
5	33583	Michael Gibbs	Brandi50@gmail.com	2566923660	972 Lopez Roads Apt. 036	Portsmouth	England	11112	UK	24	Female	Medium	Regular
6	22783	Jessica West	Laura57@gmail.com	4025769451	00526 Michael Groves	Portsmouth	England	89259	UK	24	Female	Medium	Regular
7	69900	Linda Harper	Tara9@gmail.com	3837447850	05651 Scott Squares	Portsmouth	England	93859	UK	24	Female	Medium	Regular
8	41957	Robert Williams	Laura7@gmail.com	7768491460	24163 Ashley Grove Suite 838	Portsmouth	England	96865	UK	24	Female	Medium	Regular
9	35709	Christian Rivera	Alex28@gmail.com	5647733189	710 Adrian Underpass Apt. 182	Portsmouth	England	22318	UK	24	Female	Medium	Regular
10	47148	Brian Hughes	Eric79@gmail.com	9679639442	08804 Courtney Trace Apt. 530	Portsmouth	England	1971	UK	24	Female	Medium	Regular
11	88882	Lisa Christian	Billy81@gmail.com	9569625261	72175 Nichole Mall Apt. 220	Portsmouth	England	95200	UK	24	Male	Medium	Regular
12	32448	Steven Watkins	Thomas57@gmail.com	8842167653	6364 Mcbride Vista Suite 030	Portsmouth	England	43515	UK	24	Male	Medium	Regular
13	30757	Jonathan Giles	Thomas63@gmail.com	8675357492	48252 Hunter Village Suite 716	Portsmouth	England	70486	UK	24	Male	Medium	Regular
14	52028	Chad Oconnor	Peter25@gmail.com	4915527799	67445 Ashley Well	Portsmouth	England	38239	UK	24	Female	Medium	Regular
15	21894	Kelly Carlson	Angela7@gmail.com	5908049348	597 Griffith Lake	Portsmouth	England	70281	UK	24	Female	Medium	Regular
16	97738	Mark Lee	Tammy6@gmail.com	6179852582	21409 Becky Port Apt. 193	Portsmouth	England	4683	UK	24	Female	Medium	Regular
17	91963	Kelly Howell	John43@gmail.com	8715183206	458 Michael Meadows	Portsmouth	England	60386	UK	24	Female	Medium	Regular
18	55207	Donna Mason	Travis26@gmail.com	4877455149	349 Samuel Passage	Portsmouth	England	6154	UK	24	Female	Medium	Regular
19	39540	Melissa Sharp	Stephen44@gmail.com	2007750390	8410 Christina Ford Apt. 397	Portsmouth	England	11424	UK	24	Female	Medium	Regular
20	70330	Richard Wilkins	Joseph60@gmail.com	9737167900	68431 Valerie Light	Portsmouth	England	87010	UK	24	Female	Medium	Regular
21	70761	Terri Vargas	Brenda8@gmail.com	8745207521	002 Kelley Trail Apt. 330	Portsmouth	England	11796	UK	24	Male	Medium	Regular
22	89546	Madison Fisher	David58@gmail.com	7343591276	71485 Miller Ports Apt. 413	Portsmouth	England	6368	UK	24	Female	Medium	Regular
23	67447	Paige Horton	Alyssa15@gmail.com	9113895045	42701 Michelle Overpass Apt. 240	Portsmouth	England	72959	UK	24	Male	Medium	Regular
24	99211	Daniel Dean	Jessica49@gmail.com	4714372947	6317 Diaz Fall Suite 239	Portsmouth	England	29260	UK	24	Female	Medium	Regular
25	25754	Susan Mason	Michael29@gmail.com	8027450320	42699 Krista Motorway	Portsmouth	England	23188	UK	24	Male	Medium	Regular
26	43225	Roberto Reynolds	Jeff44@gmail.com	8748515469	025 Mikayla Freeway	Portsmouth	England	79407	UK	24	Male	Medium	Regular
27	10876	Christine Smith	Diana92@gmail.com	9143770648	11977 Renton Mission	Portsmouth	England	63793	UK	24	Male	Medium	Regular

StgProductMaster

Stores extracted product master records from the Excel source.

```
CREATE TABLE dbo.StgProductMaster
(
  Product_Code      NVARCHAR(255)  NULL,
  Product_Category  NVARCHAR(50)   NULL,
  Product_Brand     NVARCHAR(50)   NULL,
  Product_Type      NVARCHAR(50)   NULL,
  ProductName       NVARCHAR(100)  NULL
);
```

	Product_Code	Product_Category	Product_Brand	Product_Type	ProductName
1	Clothing|Nike|Shorts|Cycling shorts	Clothing	Nike	Shorts	Cycling shorts
2	Electronics|Samsung|Tablet|Lenovo Tab	Electronics	Samsung	Tablet	Lenovo Tab
3	Books|Penguin Books|Children's|Sports equipment	Books	Penguin Books	Children's	Sports equipment
4	Home Decor|Home Depot|Tools|Utility knife	Home Decor	Home Depot	Tools	Utility knife
5	Grocery|Nestle|Chocolate|Chocolate cookies	Grocery	Nestle	Chocolate	Chocolate cookies
6	Electronics|Apple|Tablet|Lenovo Tab	Electronics	Apple	Tablet	Lenovo Tab
7	Electronics|Samsung|Television|QLED TV	Electronics	Samsung	Television	QLED TV
8	Clothing|Zara|Shirt|Dress shirt	Clothing	Zara	Shirt	Dress shirt
9	Grocery|Nestle|Chocolate|Dark chocolate	Grocery	Nestle	Chocolate	Dark chocolate
10	Home Decor|Home Depot|Decorations|Candles	Home Decor	Home Depot	Decorations	Candles
11	Home Decor|Home Depot|Tools|Screwdriver set	Home Decor	Home Depot	Tools	Screwdriver set
12	Books|Random House|Non-Fiction|Science	Books	Random House	Non-Fiction	Science
13	Grocery|Coca-Cola|Water|Bottled water	Grocery	Coca-Cola	Water	Bottled water
14	Grocery|Nestle|Snacks|Fruit snacks	Grocery	Nestle	Snacks	Fruit snacks
15	Clothing|Adidas|T-shirt|V-neck tee	Clothing	Adidas	T-shirt	V-neck tee
16	Books|Random House|Literature|Drama	Books	Random House	Literature	Drama
17	Grocery|Pepsi|Water|Flavored water	Grocery	Pepsi	Water	Flavored water
18	Electronics|Apple|Tablet|Samsung Galaxy Tab	Electronics	Apple	Tablet	Samsung Galaxy Tab
19	Grocery|Coca-Cola|Juice|Orange juice	Grocery	Coca-Cola	Juice	Orange juice
20	Home Decor|IKEA|Furniture|Bookshelf	Home Decor	IKEA	Furniture	Bookshelf
21	Grocery|Nestle|Coffee|Affogato	Grocery	Nestle	Coffee	Affogato
22	Books|HarperCollins|Non-Fiction|Business	Books	HarperCollins	Non-Fiction	Business
23	Books|Penguin Books|Non-Fiction|Self-help	Books	Penguin Books	Non-Fiction	Self-help
24	Home Decor|Bed Bath & Beyond|Bathroom|Sink	Home Decor	Bed Bath & Beyond	Bathroom	Sink
25	Grocery|Coca-Cola|Juice|Grape juice	Grocery	Coca-Cola	Juice	Grape juice
26	Home Decor|Bed Bath & Beyond|Kitchen|Stove	Home Decor	Bed Bath & Beyond	Kitchen	Stove

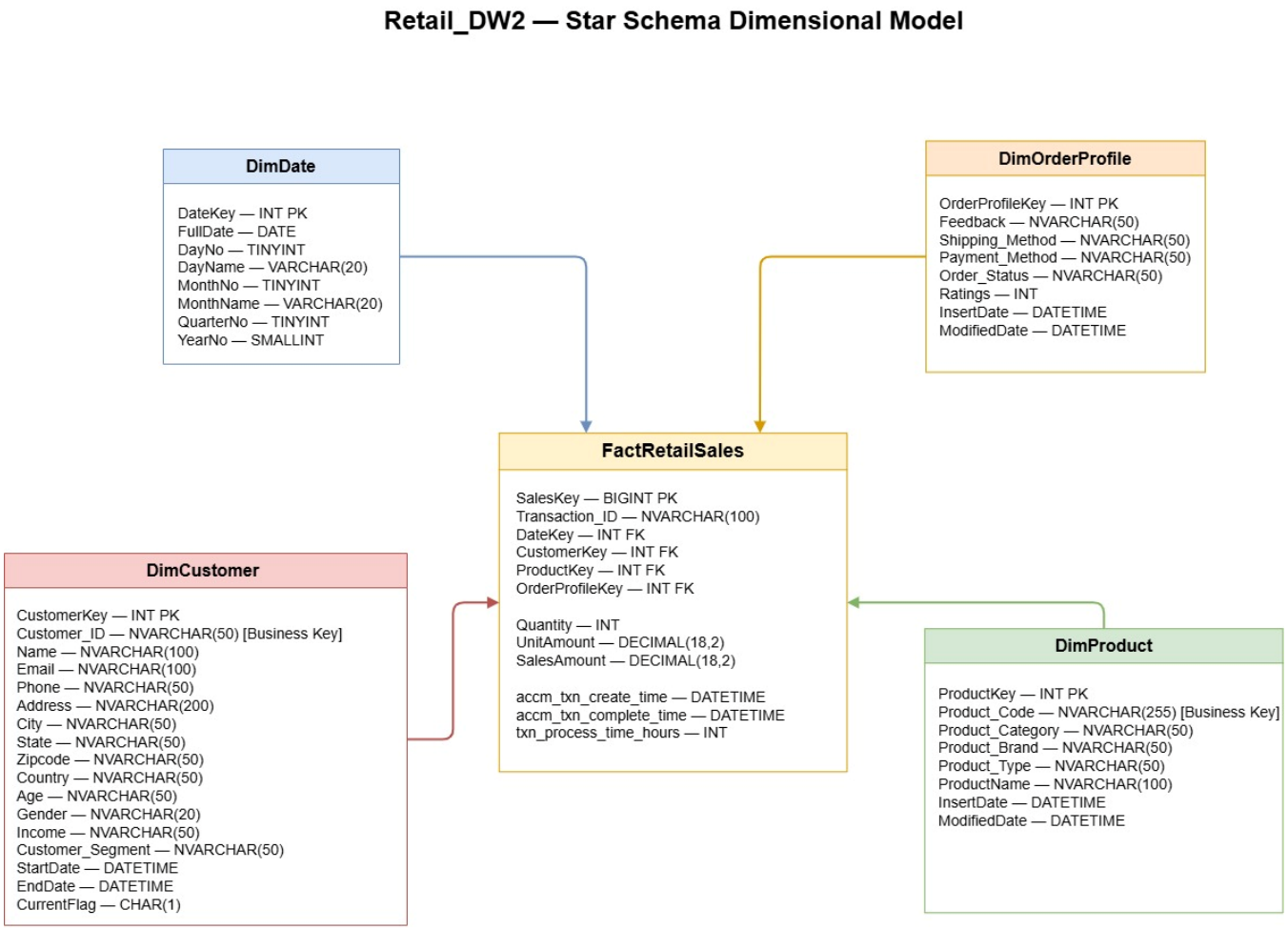
5.3 Purpose of Staging

The staging area was used to:

- isolate source extraction from warehouse loading
- perform datatype conversions
- support reloadability through truncate-before-load logic

6. Data Warehouse Design

6.1 Dimensional Model – Star Schema



6.2 Dimension Tables

	TableName	RowCount
1	DimDate	1461
2	DimCustomer	86753
3	DimProduct	592
4	DimOrderProfile	466
5	FactRetailSales	86641

DimDate

```
CREATE TABLE dbo.DimDate
(
  DateKey     INT PRIMARY KEY,
  FullDate    DATE NOT NULL UNIQUE,
  DayNo       TINYINT NOT NULL,
  DayName     VARCHAR(20) NOT NULL,
  MonthNo     TINYINT NOT NULL,
  MonthName   VARCHAR(20) NOT NULL,
  QuarterNo   TINYINT NOT NULL,
  YearNo      SMALLINT NOT NULL
);
```

	DateKey	FullDate	DayNo	DayName	MonthNo	MonthName	QuarterNo	YearNo
1	20230101	2023-01-01	1	Sunday	1	January	1	2023
2	20230102	2023-01-02	2	Monday	1	January	1	2023
3	20230103	2023-01-03	3	Tuesday	1	January	1	2023
4	20230104	2023-01-04	4	Wednesday	1	January	1	2023
5	20230105	2023-01-05	5	Thursday	1	January	1	2023
6	20230106	2023-01-06	6	Friday	1	January	1	2023
7	20230107	2023-01-07	7	Saturday	1	January	1	2023
8	20230108	2023-01-08	8	Sunday	1	January	1	2023
9	20230109	2023-01-09	9	Monday	1	January	1	2023
10	20230110	2023-01-10	10	Tuesday	1	January	1	2023
11	20230111	2023-01-11	11	Wednesday	1	January	1	2023
12	20230112	2023-01-12	12	Thursday	1	January	1	2023
13	20230113	2023-01-13	13	Friday	1	January	1	2023
14	20230114	2023-01-14	14	Saturday	1	January	1	2023
15	20230115	2023-01-15	15	Sunday	1	January	1	2023
16	20230116	2023-01-16	16	Monday	1	January	1	2023
17	20230117	2023-01-17	17	Tuesday	1	January	1	2023
18	20230118	2023-01-18	18	Wednesday	1	January	1	2023
19	20230119	2023-01-19	19	Thursday	1	January	1	2023
20	20230120	2023-01-20	20	Friday	1	January	1	2023
21	20230121	2023-01-21	21	Saturday	1	January	1	2023
22	20230122	2023-01-22	22	Sunday	1	January	1	2023
23	20230123	2023-01-23	23	Monday	1	January	1	2023
24	20230124	2023-01-24	24	Tuesday	1	January	1	2023
25	20230125	2023-01-25	25	Wednesday	1	January	1	2023
26	20230126	2023-01-26	26	Thursday	1	January	1	2023
27	20230127	2023-01-27	27	Friday	1	January	1	2023

DimProduct

```
CREATE TABLE dbo.DimProduct
(
   ProductKey        INT IDENTITY(1,1) PRIMARY KEY,
   Product_Code      NVARCHAR(255) NOT NULL,
   Product_Category  NVARCHAR(50) NULL,
   Product_Brand     NVARCHAR(50) NULL,
   Product_Type      NVARCHAR(50) NULL,
   ProductName       NVARCHAR(100) NULL,
   InsertDate        DATETIME NOT NULL DEFAULT GETDATE(),
   ModifiedDate      DATETIME NOT NULL DEFAULT GETDATE()
);
```

	ProductKey	Product_Code	Product_Category	Product_Brand	Product_Type	ProductName	InsertDate	ModifiedDate
219	219	Clothing|Adidas|T-shirt|Long-sleeve tee	Clothing	Adidas	T-shirt	Long-sleeve tee	2026-03-25 00:06:54.013	2026-03-25 00:06:54.013
220	220	Clothing|Adidas|T-shirt|Off-the-shoulder tee	Clothing	Adidas	T-shirt	Off-the-shoulder tee	2026-03-25 00:06:54.013	2026-03-25 00:06:54.013
221	221	Clothing|Adidas|T-shirt|Plain tee	Clothing	Adidas	T-shirt	Plain tee	2026-03-25 00:06:54.013	2026-03-25 00:06:54.013
222	222	Clothing|Adidas|T-shirt|Raglan tee	Clothing	Adidas	T-shirt	Raglan tee	2026-03-25 00:06:54.013	2026-03-25 00:06:54.013
223	223	Clothing|Adidas|T-shirt|Scoop neck tee	Clothing	Adidas	T-shirt	Scoop neck tee	2026-03-25 00:06:54.013	2026-03-25 00:06:54.013
224	224	Clothing|Adidas|T-shirt|V-neck tee	Clothing	Adidas	T-shirt	V-neck tee	2026-03-25 00:06:54.013	2026-03-25 00:06:54.013
225	225	Clothing|Nike|Shoes|Boots	Clothing	Nike	Shoes	Boots	2026-03-25 00:06:54.013	2026-03-25 00:06:54.013
226	226	Clothing|Nike|Shoes|Espadrilles	Clothing	Nike	Shoes	Espadrilles	2026-03-25 00:06:54.017	2026-03-25 00:06:54.017
227	227	Clothing|Nike|Shoes|Flip flops	Clothing	Nike	Shoes	Flip flops	2026-03-25 00:06:54.017	2026-03-25 00:06:54.017
228	228	Clothing|Nike|Shoes|High heels	Clothing	Nike	Shoes	High heels	2026-03-25 00:06:54.017	2026-03-25 00:06:54.017
229	229	Clothing|Nike|Shoes|Loafers	Clothing	Nike	Shoes	Loafers	2026-03-25 00:06:54.017	2026-03-25 00:06:54.017
230	230	Clothing|Nike|Shoes|Oxfords	Clothing	Nike	Shoes	Oxfords	2026-03-25 00:06:54.017	2026-03-25 00:06:54.017
231	231	Clothing|Nike|Shoes|Running shoes	Clothing	Nike	Shoes	Running shoes	2026-03-25 00:06:54.017	2026-03-25 00:06:54.017
232	232	Clothing|Nike|Shoes|Sandals	Clothing	Nike	Shoes	Sandals	2026-03-25 00:06:54.017	2026-03-25 00:06:54.017
233	233	Clothing|Nike|Shoes|Slippers	Clothing	Nike	Shoes	Slippers	2026-03-25 00:06:54.017	2026-03-25 00:06:54.017
234	234	Clothing|Nike|Shoes|Sneakers	Clothing	Nike	Shoes	Sneakers	2026-03-25 00:06:54.017	2026-03-25 00:06:54.017
235	235	Clothing|Nike|Shorts|Athletic shorts	Clothing	Nike	Shorts	Athletic shorts	2026-03-25 00:06:54.017	2026-03-25 00:06:54.017
236	236	Clothing|Nike|Shorts|Bermuda shorts	Clothing	Nike	Shorts	Bermuda shorts	2026-03-25 00:06:54.017	2026-03-25 00:06:54.017
237	237	Clothing|Nike|Shorts|Board shorts	Clothing	Nike	Shorts	Board shorts	2026-03-25 00:06:54.017	2026-03-25 00:06:54.017
238	238	Clothing|Nike|Shorts|Cargo shorts	Clothing	Nike	Shorts	Cargo shorts	2026-03-25 00:06:54.020	2026-03-25 00:06:54.020
239	239	Clothing|Nike|Shorts|Chino shorts	Clothing	Nike	Shorts	Chino shorts	2026-03-25 00:06:54.020	2026-03-25 00:06:54.020
240	240	Clothing|Nike|Shorts|Cycling shorts	Clothing	Nike	Shorts	Cycling shorts	2026-03-25 00:06:54.020	2026-03-25 00:06:54.020
241	241	Clothing|Nike|Shorts|Denim shorts	Clothing	Nike	Shorts	Denim shorts	2026-03-25 00:06:54.020	2026-03-25 00:06:54.020
242	242	Clothing|Nike|Shorts|Khaki shorts	Clothing	Nike	Shorts	Khaki shorts	2026-03-25 00:06:54.020	2026-03-25 00:06:54.020
243	243	Clothing|Nike|Shorts|Skorts	Clothing	Nike	Shorts	Skorts	2026-03-25 00:06:54.020	2026-03-25 00:06:54.020
244	244	Clothing|Nike|Shorts|Swim trunks	Clothing	Nike	Shorts	Swim trunks	2026-03-25 00:06:54.020	2026-03-25 00:06:54.020

DimOrderProfile

```
CREATE TABLE dbo.DimOrderProfile
(
   OrderProfileKey    INT IDENTITY(1,1) PRIMARY KEY,
   Feedback           NVARCHAR(50) NULL,
   Shipping_Method    NVARCHAR(50) NULL,
   Payment_Method     NVARCHAR(50) NULL,
   Order_Status       NVARCHAR(50) NULL,
   Ratings            INT NULL,
   InsertDate         DATETIME NOT NULL DEFAULT GETDATE(),
   ModifiedDate       DATETIME NOT NULL DEFAULT GETDATE()
);
```

	OrderProfileKey	Feedback	Shipping_Method	Payment_Method	Order_Status	Ratings	InsertDate	ModifiedDate
1	1	Average	Same-Day	Cash	Pending	2	2026-03-25 00:06:54.317	2026-03-25 00:06:54.317
2	2	Excellent	Same-Day	Cash	Pending	5	2026-03-25 00:06:54.327	2026-03-25 00:06:54.327
3	3	Bad	Same-Day	NULL	Shipped	1	2026-03-25 00:06:54.327	2026-03-25 00:06:54.327
4	4	Good	NULL	PayPal	Pending	3	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
5	5	Bad	Express	Credit Card	NULL	1	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
6	6	Bad	Same-Day	Credit Card	Shipped	1	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
7	7	Good	Same-Day	Credit Card	Pending	3	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
8	8	Good	Standard	PayPal	Delivered	3	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
9	9	Excellent	Standard	PayPal	Shipped	4	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
10	10	Bad	NULL	PayPal	Shipped	1	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
11	11	Good	Standard	Debit Card	Pending	4	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
12	12	Average	NULL	PayPal	Processing	2	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
13	13	Excellent	Same-Day	Debit Card	Shipped	5	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
14	14	Good	Same-Day	Debit Card	Processing	3	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
15	15	NULL	Same-Day	Debit Card	Delivered	NULL	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
16	16	Bad	Same-Day	Cash	Processing	1	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
17	17	Average	Express	Cash	Processing	2	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
18	18	Excellent	Standard	Debit Card	Shipped	4	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
19	19	NULL	Standard	PayPal	Delivered	NULL	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
20	20	Average	Express	NULL	Pending	2	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
21	21	Bad	Express	Cash	Processing	1	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
22	22	Bad	Standard	Credit Card	Processing	1	2026-03-25 00:06:54.330	2026-03-25 00:06:54.330
23	23	NULL	Same-Day	Credit Card	Delivered	NULL	2026-03-25 00:06:54.333	2026-03-25 00:06:54.333
24	24	Good	Express	NULL	Delivered	4	2026-03-25 00:06:54.333	2026-03-25 00:06:54.333
25	25	Excellent	Same-Day	Cash	NULL	4	2026-03-25 00:06:54.333	2026-03-25 00:06:54.333
26	26	Excellent	Express	Cash	Delivered	4	2026-03-25 00:06:54.333	2026-03-25 00:06:54.333
27	27	Good	Express	Debit Card	NULL	4	2026-03-25 00:06:54.333	2026-03-25 00:06:54.333

DimCustomer

```
CREATE TABLE dbo.DimCustomer
(
   CustomerKey      INT IDENTITY(1,1) PRIMARY KEY,
   Customer_ID      NVARCHAR(50) NOT NULL,
   Name             NVARCHAR(100) NULL,
   Email            NVARCHAR(100) NULL,
   Phone            NVARCHAR(50) NULL,
   Address          NVARCHAR(200) NULL,
   City             NVARCHAR(50) NULL,
   State            NVARCHAR(50) NULL,
   Zipcode          NVARCHAR(50) NULL,
   Country          NVARCHAR(50) NULL,
   Age              NVARCHAR(50) NULL,
   Gender           NVARCHAR(20) NULL,
   Income           NVARCHAR(50) NULL,
   Customer_Segment   NVARCHAR(50) NULL,
   StartDate        DATETIME NULL,
   EndDate          DATETIME NULL,
   CurrentFlag      CHAR(1) NOT NULL
);
```

	CustomerKey	Customer_ID	Name	Email	Phone	Address	City	State	Zipcode	Country	Age	Gender	Income	Customer_Segment	StartDate	EndDate	CurrentFlag
1	1	72462	Christopher Snyder	Matthew18@gmail.com	1511799368	2960 Michael Gardens Apt. 389	Portsmouth	England	18095	UK	24	Male	Medium	Regular	NULL	NULL	Y
2	2	18567	Kathryn Marshall MD	Joshua8@gmail.com	7574729332	602 Robert Locks Suite 414	Portsmouth	England	24677	UK	24	Male	Medium	Regular	NULL	NULL	Y
3	3	92909	Alex Clark	Karen49@gmail.com	2934697623	414 Cassandra Freeway Apt. 574	Portsmouth	England	98298	UK	24	Male	Medium	Regular	NULL	NULL	Y
4	4	51116	Curtis Sparks	Beth44@gmail.com	1739584671	620 Aaron Brook	Portsmouth	England	30174	UK	24	Male	Medium	Regular	NULL	NULL	Y
5	5	33583	Michael Gibbs	Brandi50@gmail.com	2566923660	972 Lopez Roads Apt. 036	Portsmouth	England	11112	UK	24	Female	Medium	Regular	NULL	NULL	Y
6	6	22783	Jessica West	Laura57@gmail.com	4025769451	00526 Michael Groves	Portsmouth	England	89259	UK	24	Female	Medium	Regular	NULL	NULL	Y
7	7	69900	Linda Harper	Tara9@gmail.com	3837447850	05651 Scott Squares	Portsmouth	England	93859	UK	24	Female	Medium	Regular	NULL	NULL	Y
8	8	41957	Robert Williams	Laura7@gmail.com	7768491460	24163 Ashley Grove Suite 838	Portsmouth	England	96865	UK	24	Female	Medium	Regular	NULL	NULL	Y
9	9	35709	Christian Rivera	Alex28@gmail.com	5647733189	710 Adrian Underpass Apt. 182	Portsmouth	England	22318	UK	24	Female	Medium	Regular	NULL	NULL	Y
10	10	47148	Brian Hughes	Eric79@gmail.com	9679639442	08804 Courtney Trace Apt. 530	Portsmouth	England	1971	UK	24	Female	Medium	Regular	NULL	NULL	Y
11	11	88882	Lisa Christian	Billy81@gmail.com	9569625261	72175 Nichole Mall Apt. 220	Portsmouth	England	95200	UK	24	Male	Medium	Regular	NULL	NULL	Y
12	12	32448	Steven Watkins	Thomas57@gmail.com	8842167653	6364 Mcbride Vista Suite 030	Portsmouth	England	43515	UK	24	Male	Medium	Regular	NULL	NULL	Y
13	13	30757	Jonathan Giles	Thomas63@gmail.com	8675357492	48252 Hunter Village Suite 716	Portsmouth	England	70486	UK	24	Male	Medium	Regular	NULL	NULL	Y
14	14	52028	Chad Oconnor	Peter25@gmail.com	4915527799	67445 Ashley Well	Portsmouth	England	38239	UK	24	Female	Medium	Regular	NULL	NULL	Y
15	15	21894	Kelly Carlson	Angela7@gmail.com	5908049348	597 Griffith Lake	Portsmouth	England	70281	UK	24	Female	Medium	Regular	NULL	NULL	Y
16	16	97738	Mark Lee	Tammy6@gmail.com	6179852582	21409 Becky Port Apt. 193	Portsmouth	England	4683	UK	24	Female	Medium	Regular	NULL	NULL	Y
17	17	91963	Kelly Howell	John43@gmail.com	8715183206	458 Michael Meadows	Portsmouth	England	60386	UK	24	Female	Medium	Regular	NULL	NULL	Y
18	18	55207	Donna Mason	Travis26@gmail.com	4877455149	349 Samuel Passage	Portsmouth	England	6154	UK	24	Female	Medium	Regular	NULL	NULL	Y
19	19	39540	Melissa Sharp	Stephen44@gmail.com	2007750390	8410 Christina Ford Apt. 397	Portsmouth	England	11424	UK	24	Female	Medium	Regular	NULL	NULL	Y
20	20	70330	Richard Wilkins	Joseph60@gmail.com	9737167900	68431 Valerie Light	Portsmouth	England	87010	UK	24	Female	Medium	Regular	NULL	NULL	Y
21	21	70761	Terri Vargas	Brenda8@gmail.com	8745207521	002 Kelley Trail Apt. 330	Portsmouth	England	11796	UK	24	Male	Medium	Regular	NULL	NULL	Y
22	22	89546	Madison Fisher	David58@gmail.com	7343591276	71485 Miller Ports Apt. 413	Portsmouth	England	6368	UK	24	Female	Medium	Regular	NULL	NULL	Y
23	23	67447	Paige Horton	Alyssa15@gmail.com	9113895045	42701 Michelle Overpass Apt. 240	Portsmouth	England	72959	UK	24	Male	Medium	Regular	NULL	NULL	Y
24	24	99211	Daniel Dean	Jessica49@gmail.com	4714372947	6317 Diaz Fall Suite 239	Portsmouth	England	29260	UK	24	Female	Medium	Regular	NULL	NULL	Y
25	25	25754	Susan Mason	Michael29@gmail.com	8027450320	42699 Krista Motorway	Portsmouth	England	23188	UK	24	Male	Medium	Regular	NULL	NULL	Y
26	26	43225	Roberto Reynolds	Jeff44@gmail.com	8748515469	025 Mikayla Freeway	Portsmouth	England	79407	UK	24	Male	Medium	Regular	NULL	NULL	Y

6.3 Fact Table

FactRetailSales

```
CREATE TABLE dbo.FactRetailSales
(
  SalesKey            BIGINT IDENTITY(1,1) PRIMARY KEY,
  Transaction_ID        NVARCHAR(50) NOT NULL,
  DateKey             INT NOT NULL,
  CustomerKey           INT NOT NULL,
  ProductKey            INT NOT NULL,
  OrderProfileKey       INT NOT NULL,
  Quantity            INT NULL,
  UnitAmount            DECIMAL(18,2) NULL,
  SalesAmount           DECIMAL(18,2) NULL,
  accm_txn_create_time    DATETIME NOT NULL,
  accm_txn_complete_time   DATETIME NULL,
  txn_process_time_hours   INT NULL,
```

```
  CONSTRAINT FK_FactRetailSales_Date FOREIGN KEY(DateKey) REFERENCES dbo.DimDate(DateKey),

  CONSTRAINT FK_FactRetailSales_Customer FOREIGN KEY(CustomerKey) REFERENCES dbo.DimCustomer(CustomerKey),

  CONSTRAINT FK_FactRetailSales_Product FOREIGN KEY(ProductKey) REFERENCES dbo.DimProduct(ProductKey),

  CONSTRAINT FK_FactRetailSales_OrderProfile FOREIGN KEY(OrderProfileKey) REFERENCES dbo.DimOrderProfile(OrderProfileKey)

);
```

	SalesKey	Transaction_ID	DateKey	CustomerKey	ProductKey	OrderProfileKey	Quantity	UnitAmount	SalesAmount	accm_txn_create_time	accm_txn_complete_time	txn_process_time_hours
5	5	4983775	20240110	40547	427	275	2	124.28	248.55	2026-03-25 00:12:45.680	2026-03-25 08:15:00.000	8
6	6	6095326	20230921	40548	322	283	4	296.29	1185.17	2026-03-25 00:12:45.680	2026-03-25 08:30:00.000	8
7	7	5434096	20230626	40549	354	94	2	315.06	630.12	2026-03-25 00:12:45.680	NULL	NULL
8	8	2344675	20230324	40550	277	16	1	46.59	46.59	2026-03-25 00:12:45.680	2026-03-25 09:00:00.000	9
9	9	4155845	20240106	40551	432	223	8	328.84	2630.71	2026-03-25 00:12:45.680	2026-03-25 09:15:00.000	9
10	10	4926148	20231004	40552	533	77	10	397.61	3976.11	2026-03-25 00:12:45.680	2026-03-25 09:30:00.000	9
11	11	8493213	20230720	40553	559	309	4	90.98	363.93	2026-03-25 00:12:45.680	2026-03-25 09:45:00.000	9
12	12	1609659	20230621	40554	182	456	1	364.83	364.83	2026-03-25 00:12:45.680	2026-03-25 10:00:00.000	10
13	13	2846832	20240102	40555	418	419	10	161.88	1618.79	2026-03-25 00:12:45.680	2026-03-25 10:15:00.000	10
14	14	3217461	20230507	40556	450	347	9	350.82	3157.35	2026-03-25 00:12:45.680	2026-03-25 10:30:00.000	10
15	15	2401331	20231118	40557	224	80	6	297.73	1786.36	2026-03-25 00:12:45.680	2026-03-25 10:45:00.000	10
16	16	4969987	20230615	40558	168	337	3	23.72	71.17	2026-03-25 00:12:45.680	2026-03-25 11:00:00.000	11
17	17	6681000	20230701	40559	481	300	10	14.73	147.27	2026-03-25 00:12:45.680	2026-03-25 11:15:00.000	11
18	18	2430572	20230414	40560	324	377	3	139.79	419.36	2026-03-25 00:12:45.680	2026-03-25 11:30:00.000	11
19	19	7728099	20240207	40561	402	388	1	210.14	210.14	2026-03-25 00:12:45.680	2026-03-25 11:45:00.000	11
20	20	3489336	20231024	40562	574	282	2	469.66	939.32	2026-03-25 00:12:45.680	2026-03-25 12:00:00.000	12
21	21	6925572	20230321	40563	436	93	6	420.88	2525.25	2026-03-25 00:12:45.680	2026-03-25 12:15:00.000	12
22	22	8871554	20230806	40564	106	293	4	200.29	801.15	2026-03-25 00:12:45.680	2026-03-25 12:30:00.000	12
23	23	9476855	20240203	40565	153	93	6	34.12	204.75	2026-03-25 00:12:45.680	2026-03-25 12:45:00.000	12
24	24	9338306	20230531	40566	507	176	8	436.61	3492.86	2026-03-25 00:12:45.680	2026-03-25 13:00:00.000	13
25	25	9070447	20240227	40567	398	192	2	285.30	570.60	2026-03-25 00:12:45.680	2026-03-25 13:15:00.000	13
26	26	9082305	20230420	40568	531	51	6	75.21	451.26	2026-03-25 00:12:45.680	2026-03-25 13:30:00.000	13
27	27	4725198	20230603	40569	327	338	1	173.57	173.57	2026-03-25 00:12:45.680	2026-03-25 13:45:00.000	13
28	28	5230876	20240107	40570	317	6	5	471.48	2357.41	2026-03-25 00:12:45.680	2026-03-25 14:00:00.000	14
29	29	2459574	20230422	40571	535	204	8	333.86	2670.88	2026-03-25 00:12:45.680	2026-03-25 14:15:00.000	14
30	30	2873384	20230621	40572	404	79	6	240.23	1441.37	2026-03-25 00:12:45.680	2026-03-25 14:30:00.000	14

6.4 Design Assumptions

DimDate is pre-loaded with the date range 2023-2026 to ensure all valid dates for the transaction can be mapped to a DateKey, allowing for some extension into the future outside the current source date range.

The original retail data source did not include a numeric product identifier, so Product_Code was created as part of the source preparation by concatenating Product_Category, Product_Brand, Product_Type, and ProductName.

The original data source did not include a separate natural key for order profile attributes, so distinct combinations of Feedback, Shipping_Method, Payment_Method, Order_Status, and Ratings were used as unique order profiles, each of which was assigned a surrogate key OrderProfileKey in DimOrderProfile.

To avoid potential datatype errors between SQL Server, CSV, Excel, and SSIS metadata, Customer_ID, Transaction_ID, and Product_Code were all processed as text (NVARCHAR).

Rows in StgRetailOrders with NULL Date values are excluded from fact loading, because those rows cannot be assigned a valid DateKey through the date lookup.

Update rows where Transaction_ID does not match an existing fact row, treated as unmatched update records and redirected to a reject file (txn_update_rejects.csv).

7. ETL Package Implementation

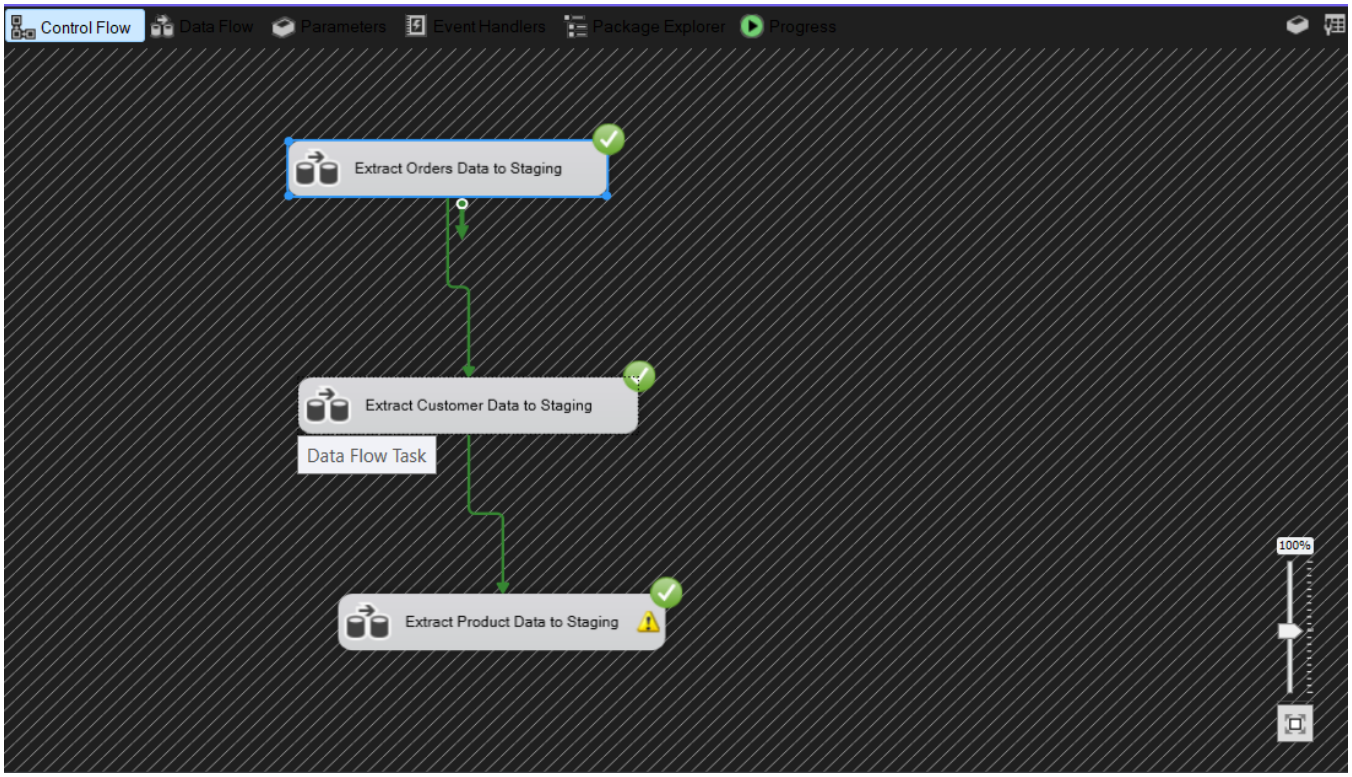
Three SSIS packages were implemented.

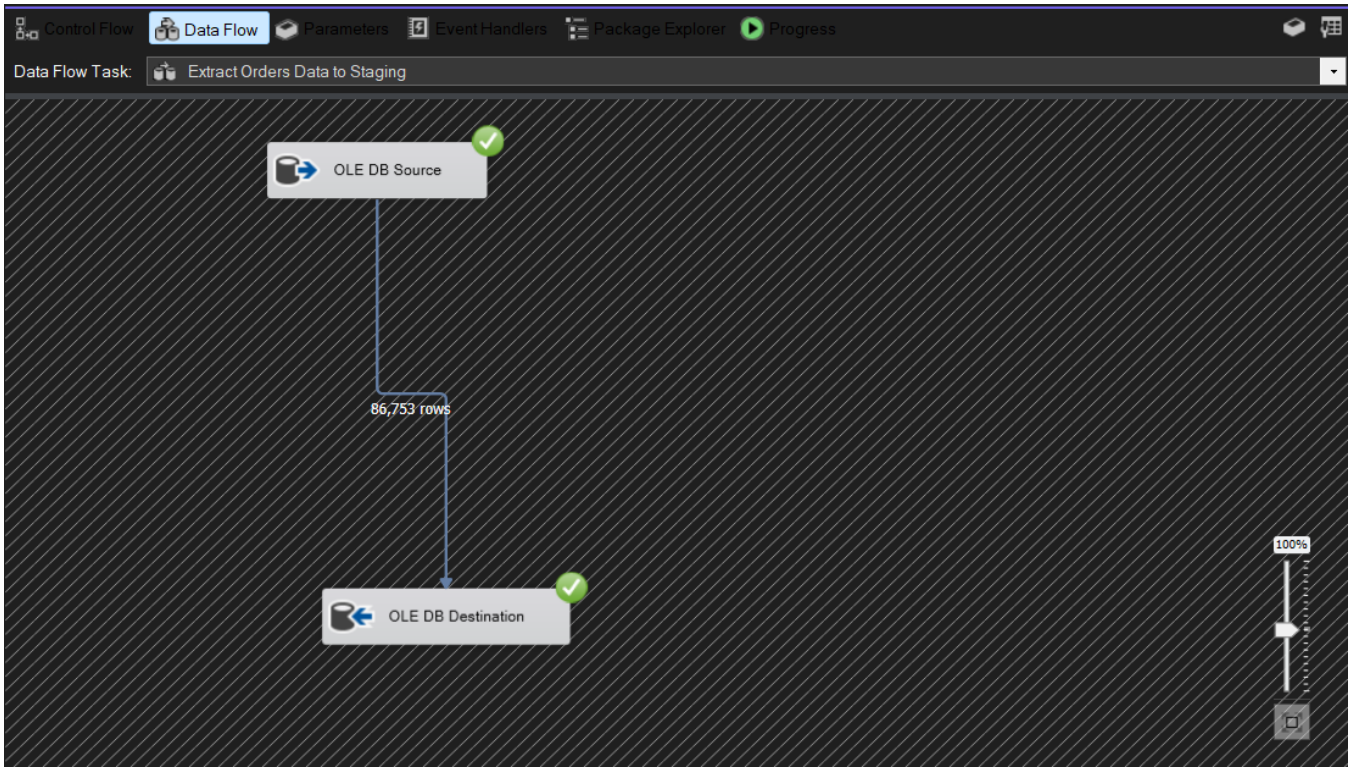
7.1 Package 1 – Retail_Load_Staging.dtsx

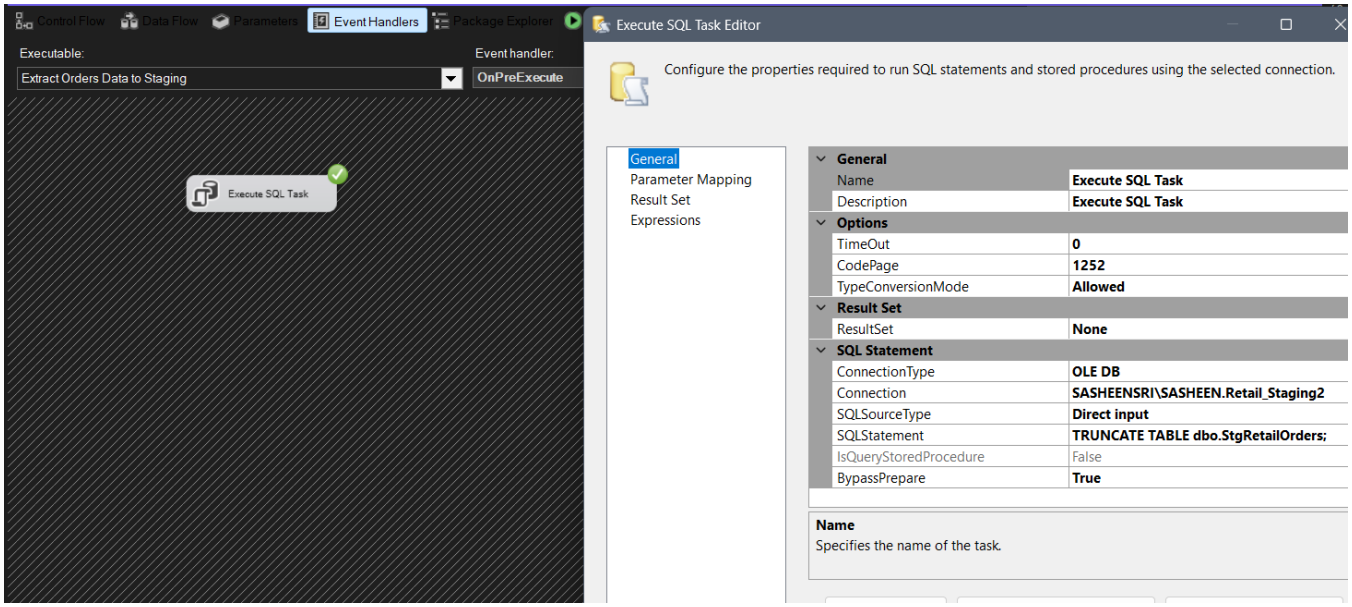
This package extracts data from the prepared sources and loads the staging tables.

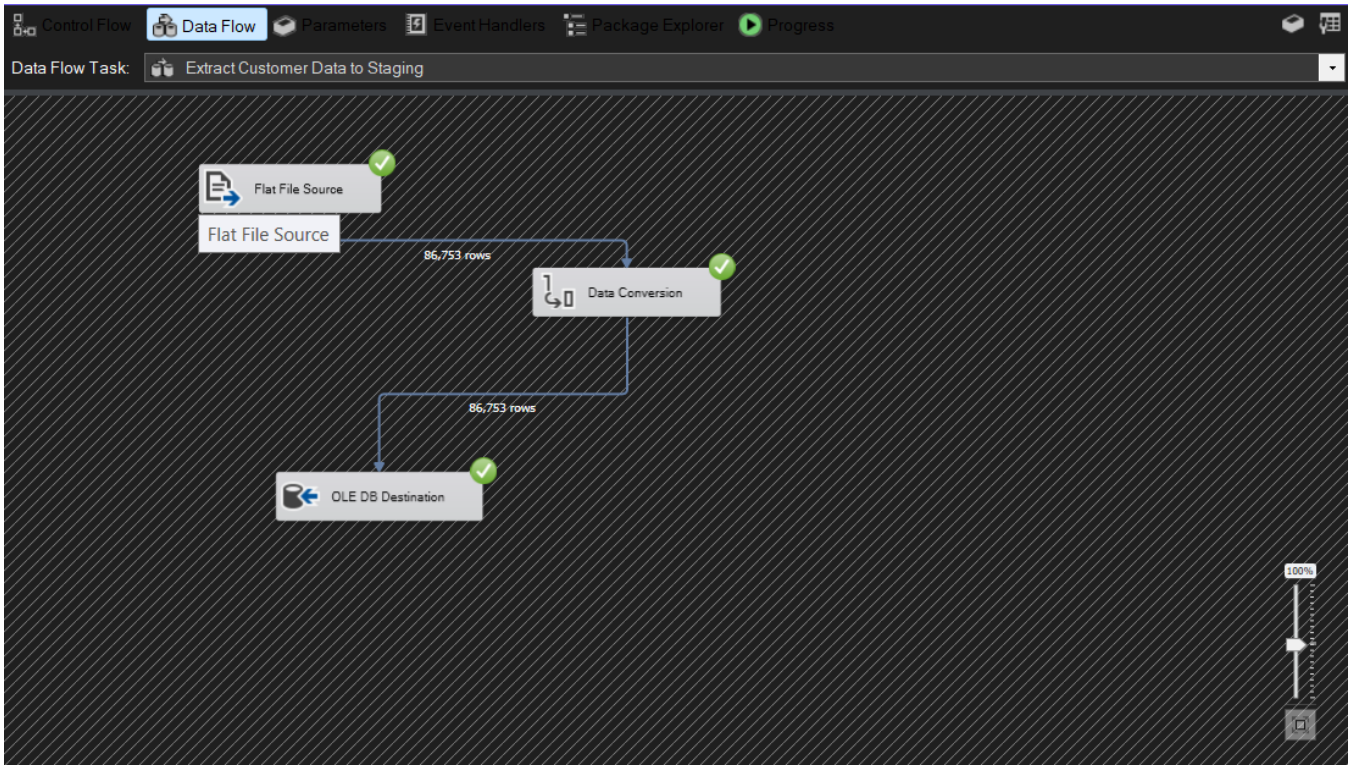
Tasks performed:

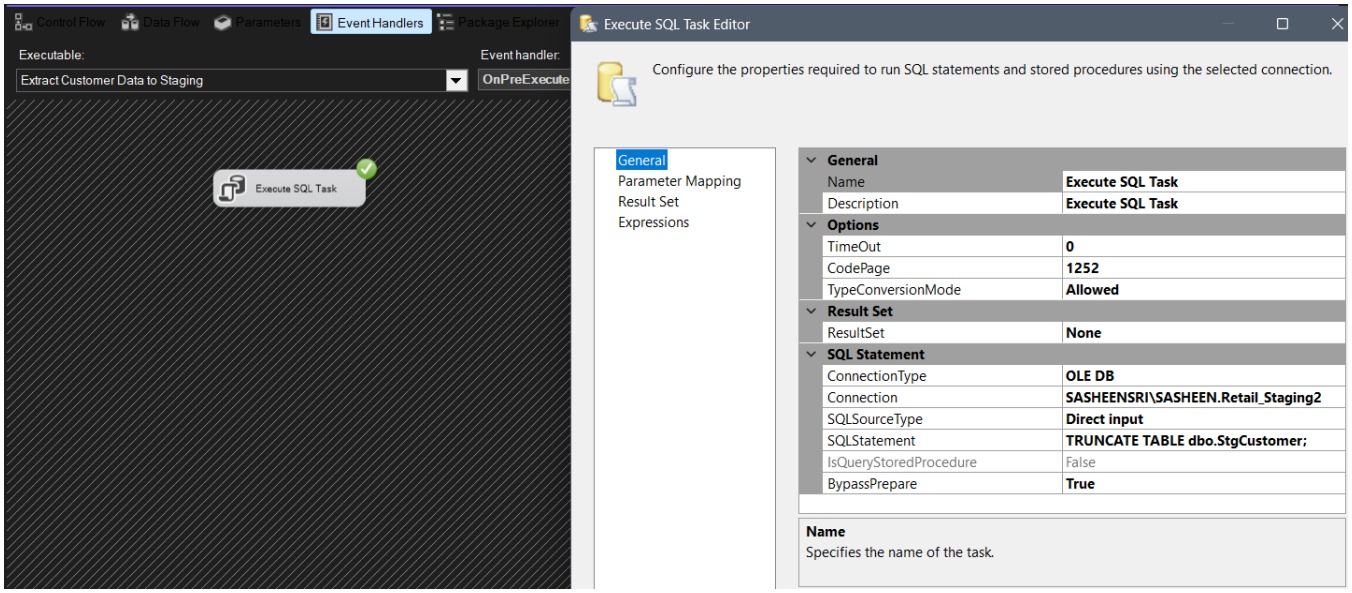
- extract order data from Retail_SourceDB2.dbo.SrcRetailOrders
- load order data into Retail_Staging2.dbo.StgRetailOrders
- extract customer data from Customer_Master.csv
- load customer data into Retail_Staging2.dbo.StgCustomer
- extract product data from Product_Master.xlsx
- load product data into Retail_Staging2.dbo.StgProductMaster
- truncate staging tables before each load using **OnPreExecute Event Handlers**

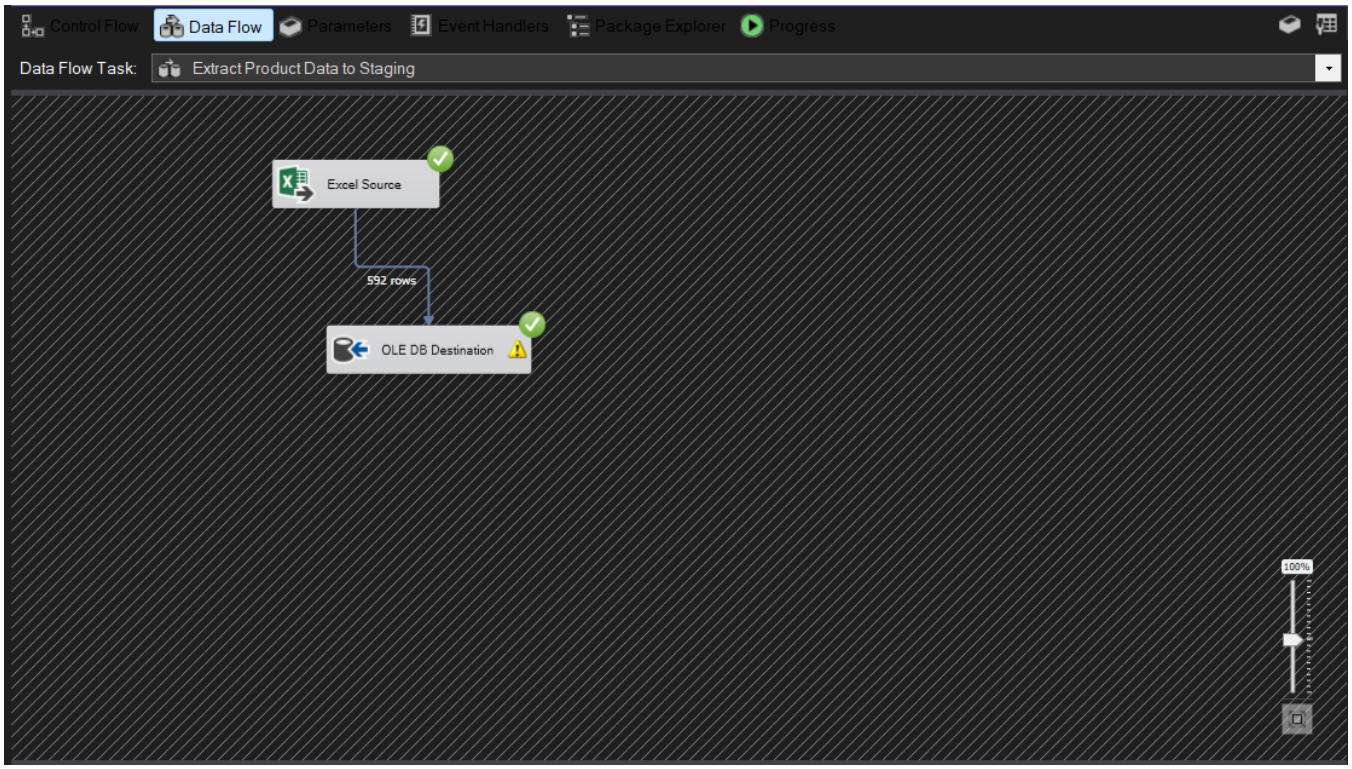

Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Extract Orders Data to Staging
Extract Customer Data to Staging
Data Flow Task
Extract Product Data to Staging
100%

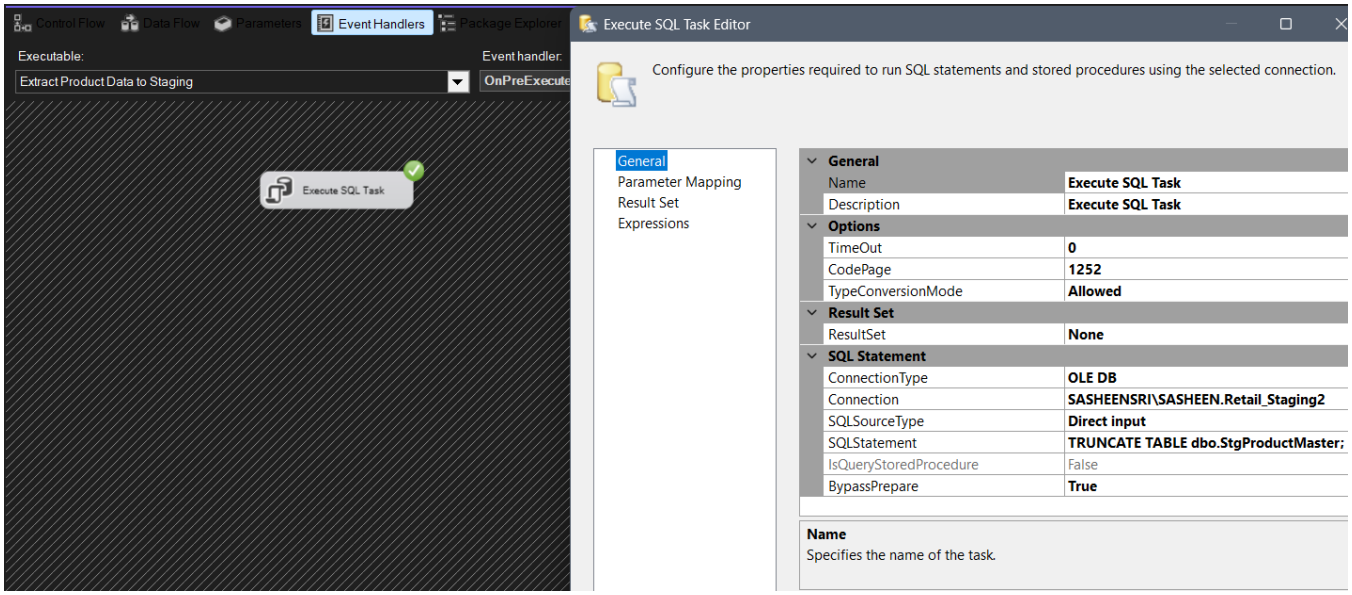

Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
Extract Orders Data to Staging
OLE DB Source
86,753 rows
OLE DB Destination
100%


Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Executable:
Extract Orders Data to Staging
Event handler:
OnPreExecute
Execute SQL Task
Execute SQL Task Editor
Configure the properties required to run SQL statements and stored procedures using the selected connection.
General
Parameter Mapping
Result Set
Expressions
General
Name
Execute SQL Task
Description
Execute SQL Task
Options
TimeOut
0
CodePage
1252
TypeConversionMode
Allowed
Result Set
ResultSet
None
SQL Statement
ConnectionType
OLE DB
Connection
SASHEENSRI\SASHEEN.Retail_Staging2
SQLSourceType
Direct input
SQLStatement
TRUNCATE TABLE dbo.StgRetailOrders;
IsQueryStoredProcedure
False
BypassPrepare
True
Name
Specifies the name of the task.


Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
Extract Customer Data to Staging
Flat File Source
Flat File Source
86,753 rows
Data Conversion
86,753 rows
OLE DB Destination
100%

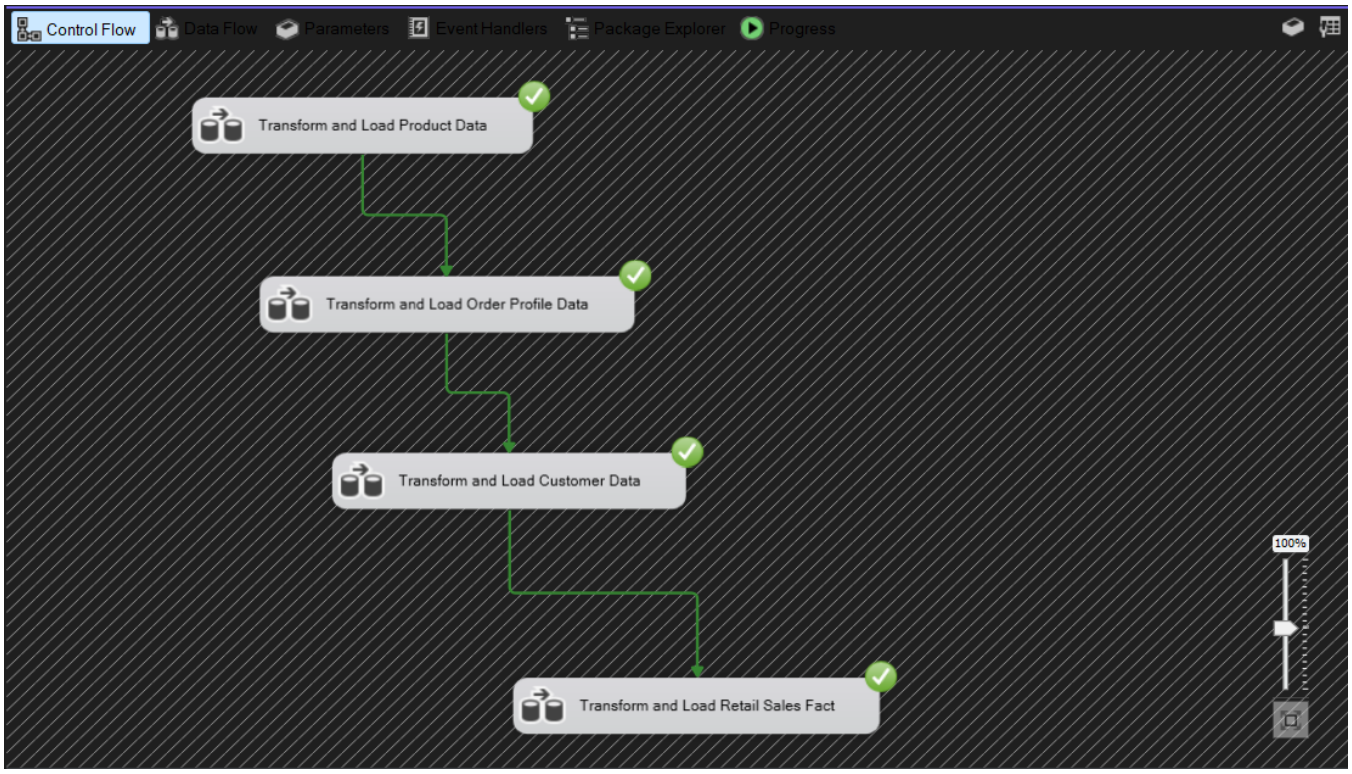

Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Executable:
Extract Customer Data to Staging
Event handler:
OnPreExecute
Execute SQL Task
Execute SQL Task Editor
Configure the properties required to run SQL statements and stored procedures using the selected connection.
General
Parameter Mapping
Result Set
Expressions
General
Name
Execute SQL Task
Description
Execute SQL Task
Options
TimeOut
0
CodePage
1252
TypeConversionMode
Allowed
Result Set
ResultSet
None
SQL Statement
ConnectionType
OLE DB
Connection
SASHEENSRI\SASHEEN.Retail_Staging2
SQLSourceType
Direct input
SQLStatement
TRUNCATE TABLE dbo.StgCustomer;
IsQueryStoredProcedure
False
BypassPrepare
True
Name
Specifies the name of the task.

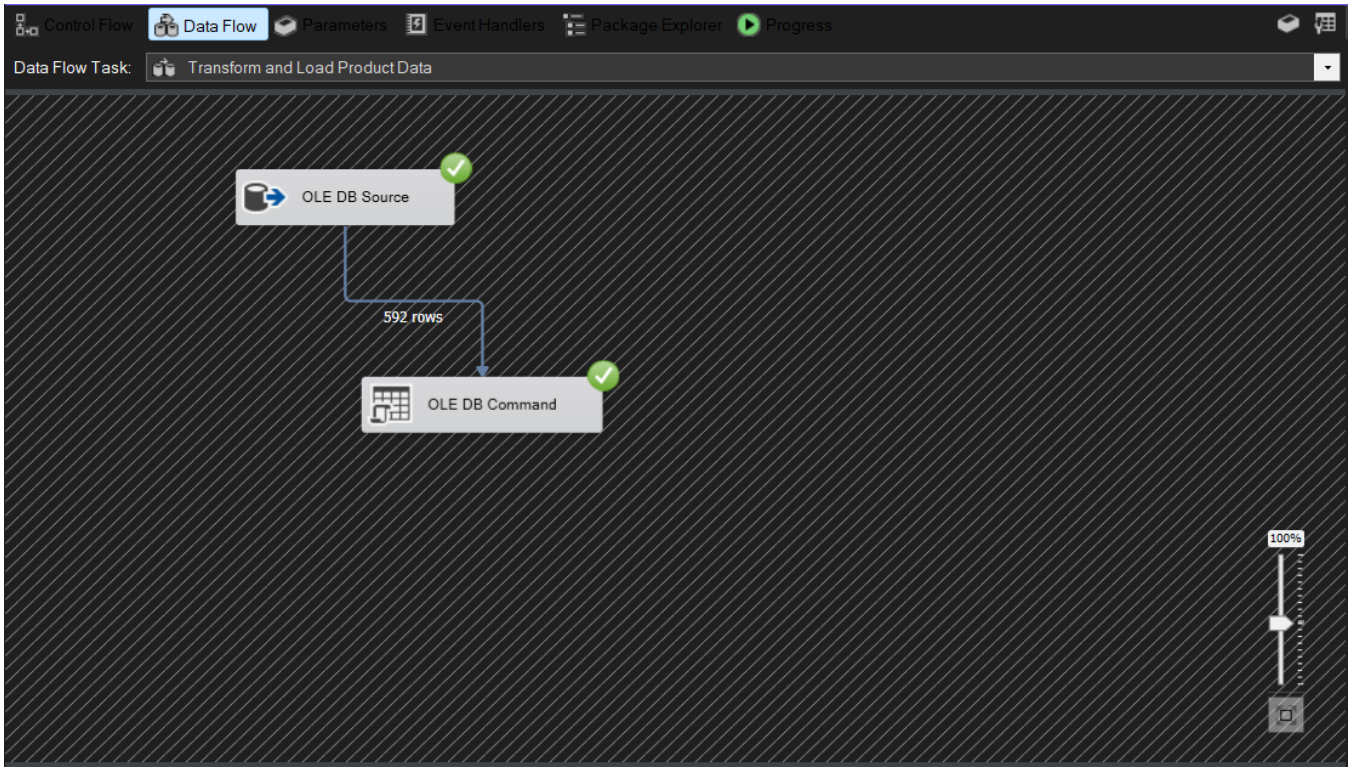

Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
Extract Product Data to Staging
Excel Source
592 rows
OLE DB Destination
100%

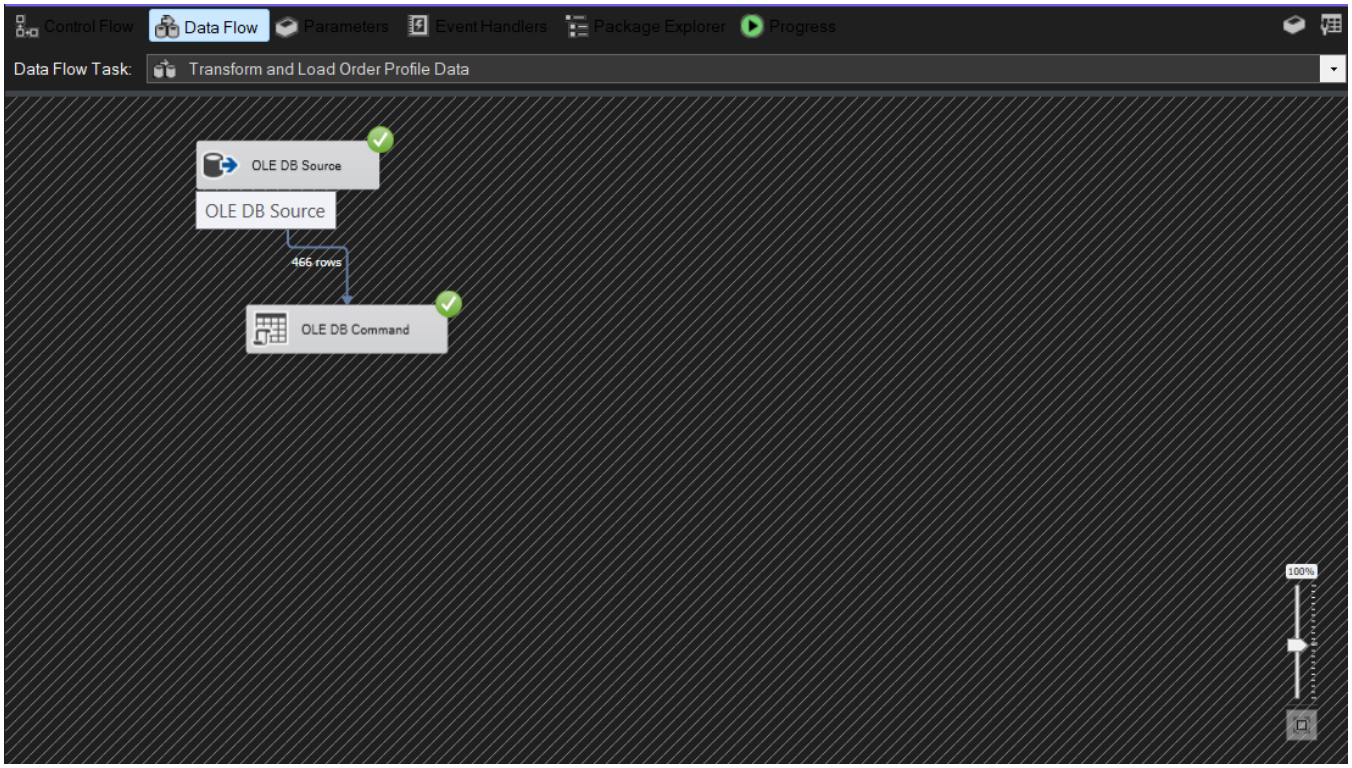


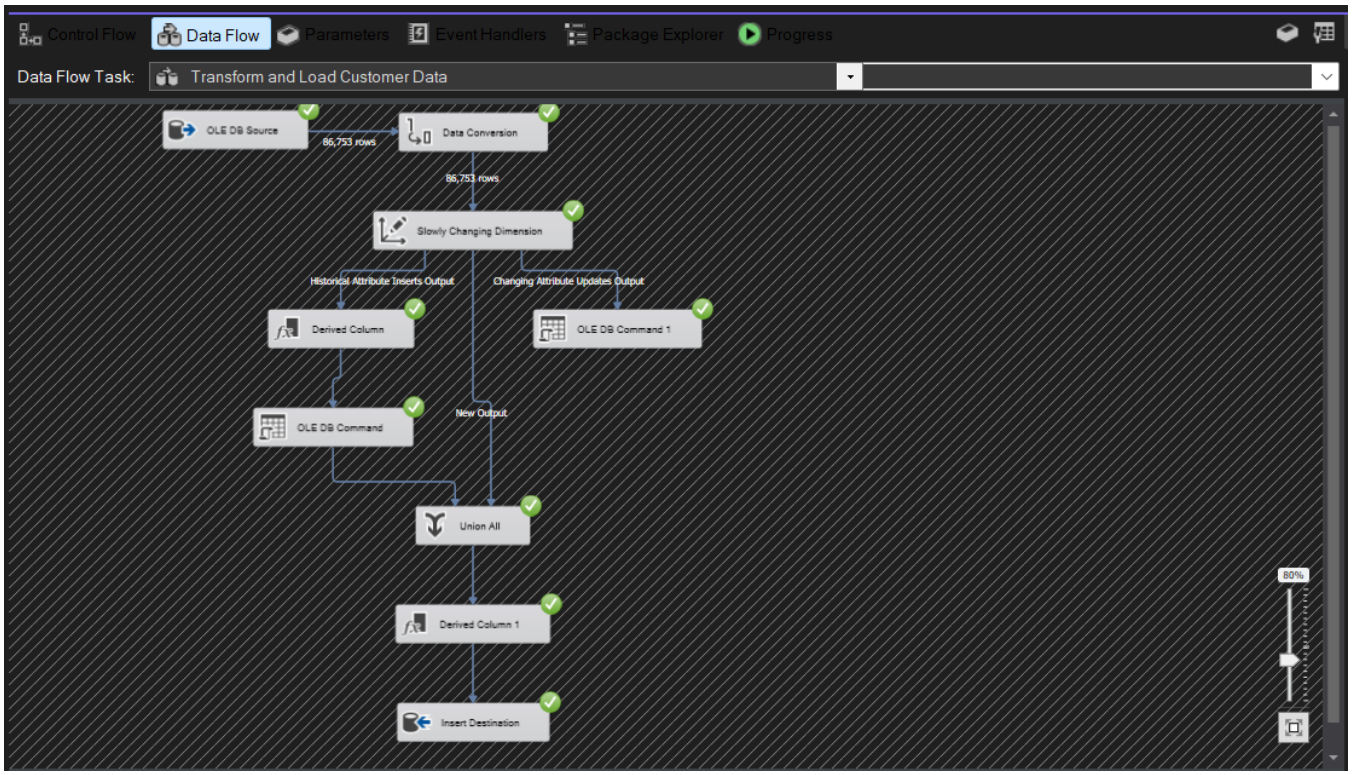
7.2 Package 2 – Retail_Load_DW.dtsx

This package transforms staging data and loads the warehouse.




Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
Transform and Load Product Data
OLE DB Source
592 rows
OLE DB Command
100%


Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
Transform and Load Order Profile Data
OLE DB Source
OLE DB Source
466 rows
OLE DB Command
100%



The **customer dimension** implemented as a **Slowly Changing Dimension** using the SSIS Slowly Changing Dimension Wizard.

Business Key

- Customer_ID

Historical Attributes

- Address
- City
- State
- Zipcode
- Country
- Customer_Segment

Changing Attributes

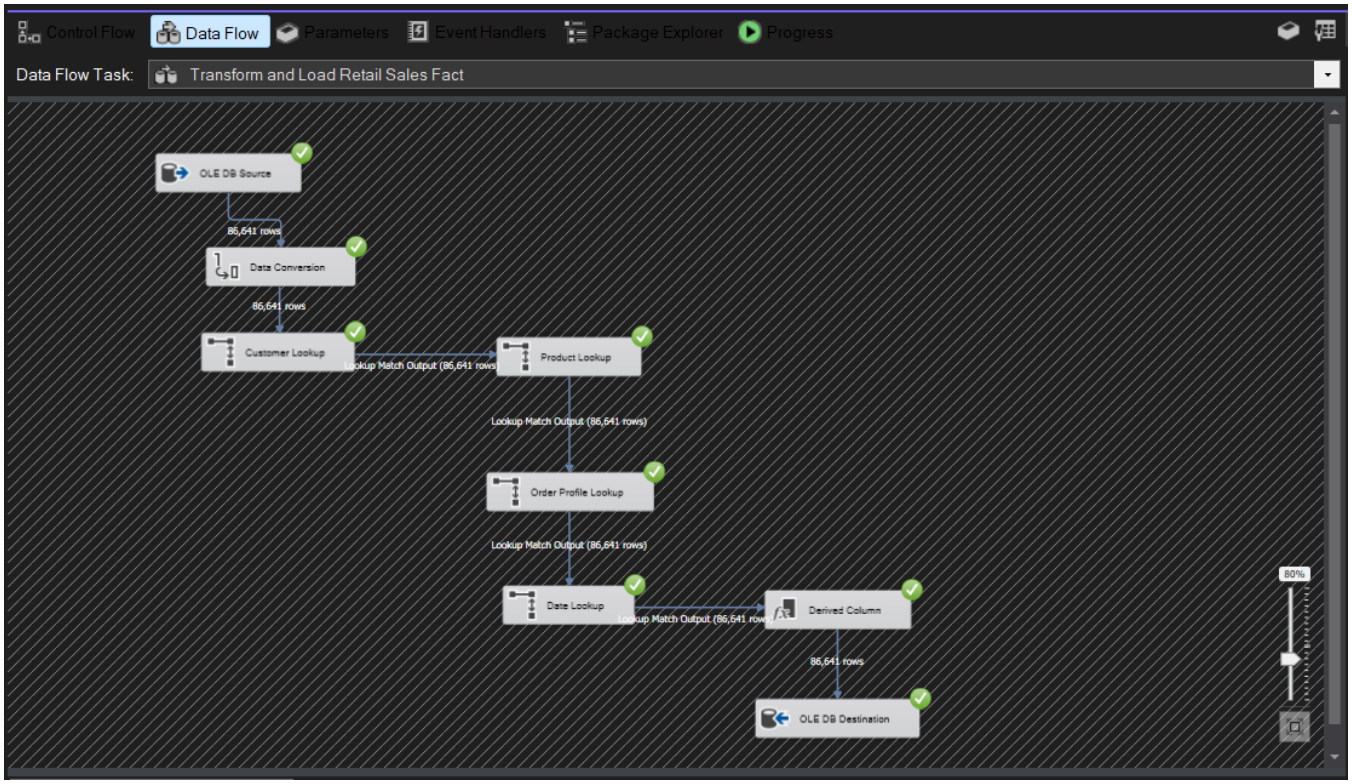
- Name
- Email
- Phone
- Age
- Gender
- Income

Current Row Indicator

The SCD logic uses:

- CurrentFlag = 'Y' for current rows
- CurrentFlag = 'N' for historical rows

This allows current customer records to be separated from expired versions.



Stored Procedures Used in Dimension Loading

Two stored procedures, dbo.UpdateDimProduct and dbo.UpdateDimOrderProfile, were created in SSMS and called from SSIS using OLE DB Command during Retail_Load_DW.dtsx. They were used to apply insert/update logic for DimProduct and DimOrderProfile during warehouse loading.

```sql
CREATE OR ALTER PROCEDURE dbo.UpdateDimProduct
    @Product_Code NVARCHAR(255),
    @Product_Category NVARCHAR(50),
    @Product_Brand NVARCHAR(50),
    @Product_Type NVARCHAR(50),
    @ProductName NVARCHAR(100)
AS
BEGIN
    SET NOCOUNT ON;

    IF NOT EXISTS
    (
        SELECT 1
        FROM dbo.DimProduct
        WHERE Product_Code = @Product_Code
    )
    BEGIN
        INSERT INTO dbo.DimProduct
        (Product_Code, Product_Category, Product_Brand, Product_Type, ProductName, InsertDate, ModifiedDate)
        VALUES
        (@Product_Code, @Product_Category, @Product_Brand, @Product_Type, @ProductName, GETDATE(), GETDATE());
    END
    ELSE
    BEGIN
        UPDATE dbo.DimProduct
        SET
            Product_Category = @Product_Category,
            Product_Brand = @Product_Brand,
            Product_Type = @Product_Type,
            ProductName = @ProductName,
            ModifiedDate = GETDATE()
        WHERE Product_Code = @Product_Code;
    END
END;
```

```sql
CREATE OR ALTER PROCEDURE dbo.UpdateDimOrderProfile
    @Feedback NVARCHAR(50),
    @Shipping_Method NVARCHAR(50),
    @Payment_Method NVARCHAR(50),
    @Order_Status NVARCHAR(50),
    @Ratings INT
AS
BEGIN
    SET NOCOUNT ON;

    IF NOT EXISTS
    (
        SELECT 1
        FROM dbo.DimOrderProfile
        WHERE ISNULL(Feedback,'') = ISNULL(@Feedback,'')
          AND ISNULL(Shipping_Method,'') = ISNULL(@Shipping_Method,'')
          AND ISNULL(Payment_Method,'') = ISNULL(@Payment_Method,'')
          AND ISNULL(Order_Status,'') = ISNULL(@Order_Status,'')
          AND ISNULL(Ratings,-1) = ISNULL(@Ratings,-1)
    )
    BEGIN
        INSERT INTO dbo.DimOrderProfile
        (Feedback, Shipping_Method, Payment_Method, Order_Status, Ratings, InsertDate, ModifiedDate)
        VALUES
        (@Feedback, @Shipping_Method, @Payment_Method, @Order_Status, @Ratings, GETDATE(), GETDATE());
    END
    ELSE
    BEGIN
        UPDATE dbo.DimOrderProfile
        SET ModifiedDate = GETDATE()
        WHERE ISNULL(Feedback,'') = ISNULL(@Feedback,'')
          AND ISNULL(Shipping_Method,'') = ISNULL(@Shipping_Method,'')
          AND ISNULL(Payment_Method,'') = ISNULL(@Payment_Method,'')
          AND ISNULL(Order_Status,'') = ISNULL(@Order_Status,'')
          AND ISNULL(Ratings,-1) = ISNULL(@Ratings,-1);
    END
END;
```

7.3 Package 3 – Retail_Update_AccumulatingFact.dtsx

This package updates the accumulating fact table using a separate CSV update file:

txn_completion_updates.csv

Tasks performed:

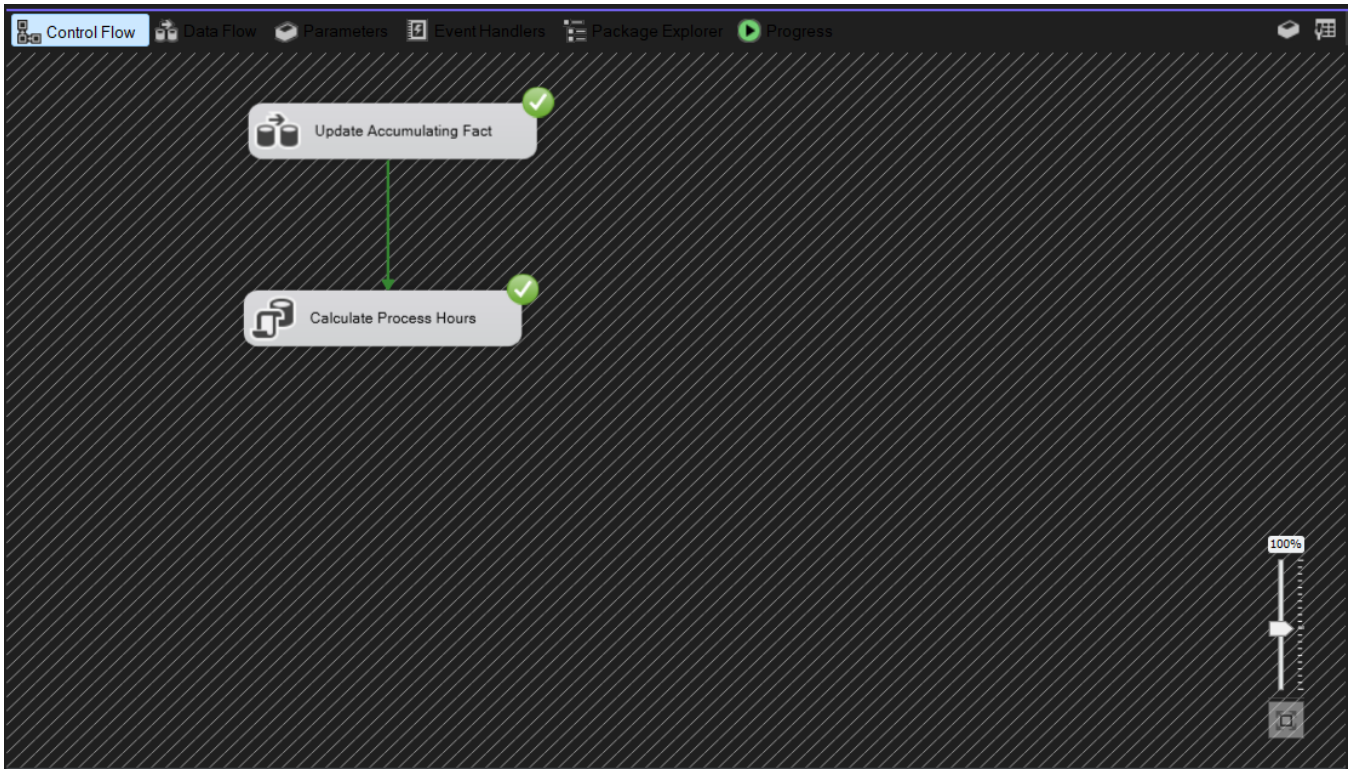
- extract completion timestamps from CSV
- match transaction rows using lookup
- update accm_txn_complete_time
- redirect unmatched update rows to reject output
- calculate txn_process_time_hours

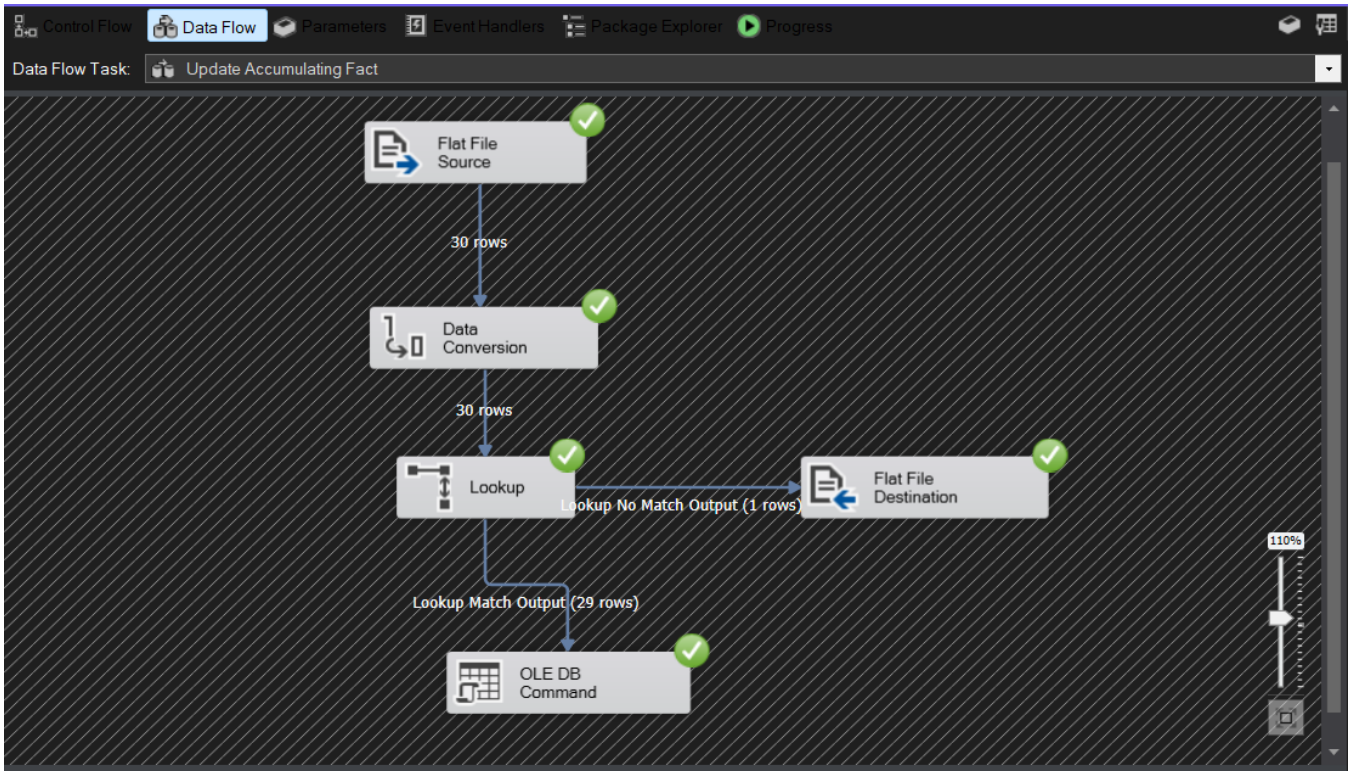
The fact table was loaded after all dimensions were populated. Lookups were used to retrieve:

- CustomerKey
- ProductKey
- OrderProfileKey
- DateKey

The fact measures loaded were:

- Quantity
- UnitAmount
- SalesAmount


Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Update Accumulating Fact
Calculate Process Hours
100%


Control Flow
Data Flow
Parameters
Event Handlers
Package Explorer
Progress
Data Flow Task:
Update Accumulating Fact
Flat File Source
30 rows
Data Conversion
30 rows
Lookup
Lookup No Match Output (1 rows)
Flat File Destination
Lookup Match Output (29 rows)
OLE DB Command
110%

8. Data Quality Issues and Resolutions

Several practical ETL issues were encountered and resolved during the implementation.

1. **Null Dates in Sources.**

Null dates were present in some of the order records, which resulted in the failure of the date lookup.

Resolution:

These records were removed in the fact source query before the data was loaded into the FactRetailSales table.

2. **Datatype Mismatches**

Columns such as Customer_ID, Transaction_ID, and Age resulted in datatype mismatch issues.

Resolution:

Data Conversion transformations were used to resolve the datatype mismatch issues.

3. **Unmatched Update Rows**

Unmatched records were present in the update file, which were not found in the fact table.

Resolution:

The no-match records were sent to the reject file.